

Literature Handbook — v2

In-depth study notes for Chapter 2 of ‘This Time Is Different?’

Updated for the three-market design (ISEQ, S&P 500, DAX; FTSE for robustness). 61 paper cards with 12 fields each (summary, dataset, methodology, results, relation to your study, exact pages to read, what to skip, key terms, strengths, weaknesses, how to use) · 11 original figures from your own data · key numbers · Irish timeline · concept primers · 24 September 2026, updated 30 September 2026 (5 new cards and 3 figures for the exploratory extensions and the closest new work; Gallagher & Twomey correction), updated 1 October 2026 (10 new cards in D6, primer C7 and 2 figures for the exploratory extensions E3–E4)

Why there are no screenshots: No paragraphs or page images are copied from the papers. They are copyrighted, and copied text in your dissertation would be plagiarism. Instead, each card tells you exactly which section, pages, tables and figures to read in the original (field ‘Must read in the original’), and the figures in this handbook are original charts made from YOUR dataset that illustrate what each key paper describes. You may reuse these figures in your dissertation because they are your own analysis — describe them in your own words.

Card colour	Meaning
Green	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Yellow	ABSTRACT / summary read in this session; read the full paper before relying on details
Grey	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Blue rows	The rows that matter most for you: relation to your study, what to read, and how to use it

Contents: A. Key numbers · B. Irish timeline · C. Concept primers with figures · D. Paper cards (2.2 theory, 2.3 crises & Ireland, 2.4 transmission, 2.5 forecasting, 2.6 explainability, methods of E3–E4) · E. Reading order · F. From notes to paragraph · G. Where to read the originals.

A. Key numbers to remember

Fact	Number	Source
ISEQ pre-GFC peak → trough	9,981 (20 Feb 2007) → 1,916 (9 Mar 2009): –80.8%	Your data
ISEQ 29 Sep 2008 / 30 Sep 2008 (guarantee day)	–13.0% / +7.9%	Your data
Irish bank CDS on guarantee day	≈400 → ≈150 bp overnight; Irish sovereign CDS rose sharply	Acharya et al. (2014), p.3
European average CDS bank / sovereign	63/14 bp before bailouts; 148/48 during; 184/112 after	Acharya et al. (2014)
Irish lending growth 1997→2008	mortgages ≈7×; developer loans ≈11×	Kelly (2009)
EU/IMF programme (28 Nov 2010)	€85bn; €67.5bn external (€45bn EU & bilateral, €22.5bn IMF)	European Commission
Programme exit	15 Dec 2013	European Commission
Brexit: ISEQ 24 / 27 Jun 2016	–7.7% / –9.9%	Your data
COVID: ISEQ 12 Mar 2020	–9.9%	Your data
War: ISEQ 24 Feb 2022	–4.6%	Your data
US daily moves >2.5%, 1900–2019	>1,100, none attributed to pandemics	Baker et al. (2020)
COVID-attributed jumps 24 Feb–20 Apr 2020	≈ two dozen	Baker et al. (2020)
COVID excess mortality vs Spanish Flu	≈1/25th	Baker et al. (2020)
Bekaert et al. sample	415 country-sector portfolios, 55 markets, ≈725 weekly observations	Bekaert et al. (2014)
Diebold–Yilmaz sample	19 markets, weekly, Jan 1992 – Nov 2007; VAR(2); 10-week horizon; 200-week rolling window	Diebold & Yilmaz (2009)
Christensen et al. sample	DJIA stocks, 29 Jan 2001 – 31 Dec 2017, 4,257 days	Christensen et al. (2023)
Bucci sample	monthly S&P 500, Feb 1950 – Dec 2017 (815 obs.); 245 out-of-sample forecasts, only 22 in the 2007–09 subsample	Bucci (2020)

Akhtaruzzaman et al. periods	pre-COVID 1 Jan 2013 – 30 Dec 2019; COVID 31 Dec 2019 – 20 Mar 2020	Akhtaruzzaman et al. (2021)
ISEQ 20 value lost 2023–24	≈€74bn of ≈€152bn (CRH, Flutter, Smurfit)	Irish Times
Your main sample	5,243 complete days, Jan 2003 – Aug 2023	Your data
Diebold–Yilmaz (2012) set-up	daily range volatility, 4 US markets, 25 Jan 1999 – 29 Jan 2010 (2,771 days); VAR(4); 10-day horizon; 200-day windows; total spillover 12.6%	Diebold & Yilmaz (2012), pp.59–61
Danielsson et al. (2018) database	60 countries, up to 211 years, 3,700 country-years, 259 banking crises; 1% below trend for 5 years → +1.01% crisis probability	Danielsson et al. (2018), pp.5–6
Your E1 (exploratory)	total spillover 34.0%; ISEQ net receiver –8.4; SHAP vs Diebold–Yilmaz agree in 4 of 8 direction checks	Your data
Your E2 (exploratory)	pre-GFC below-trend measure: Ireland 40th percentile (history 1970–2024), US 42nd, Germany 4th	Your data (FRED)
Your E3 (exploratory)	all-days QLIKE: Random Forest 0.456, SVR 0.457 vs GARCH 0.380; both under-forecast on 69% of GFC days; 2008 peak forecasts 21–23 vs actual 292 ($\times 10^{-4}$)	Your data
Your E4 (exploratory)	GARCH hybrids 0.376 / 0.376 / 0.394 vs GARCH 0.380 (Holm $p \geq 0.36$); their GFC QLIKE 0.36–0.46; HAR-X 0.361 overall but forecasts up to $9,955 \times 10^{-4}$ in October 2008	Your data
Branco et al. (2024) set-up	10 global stock indices, Jan 2000 – Dec 2021; ML under-forecasts in high-volatility periods	Branco et al. (2024), abstract
Audrino & Chassot (2024) set-up	1,445 US stocks, 2016 – Nov 2023; HAR (WLS) re-estimated daily beats every ML model	Audrino & Chassot (2024), Sections 3–5

How six shocks reached the Irish market (your data, Yahoo Finance closes)

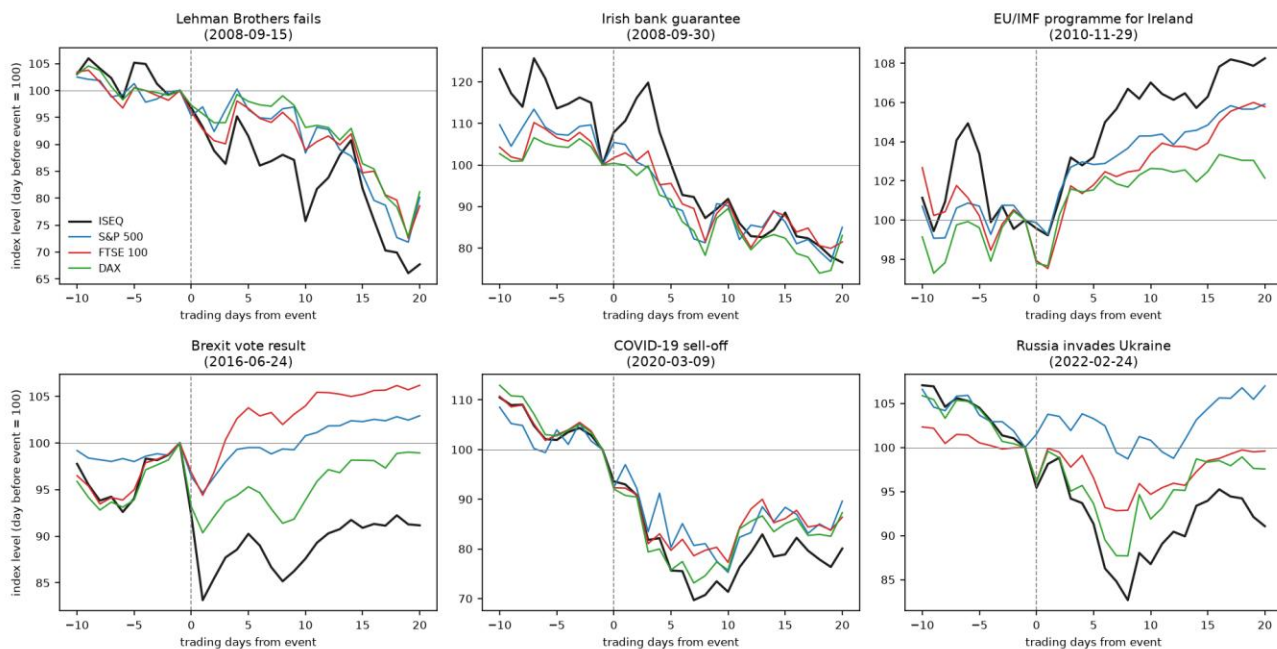


Figure H1. Your data: ISEQ, S&P 500, FTSE 100 and DAX around six events (index = 100 on the day before). Things to notice: after Lehman the ISEQ fell further than the others (domestic weakness — Bekaert et al.); on guarantee day the ISEQ jumped while others did not, then fell with them (Acharya et al.); after Brexit the ISEQ fell hardest while the FTSE 100 recovered (it is priced in sterling, which fell); in 2022 the ISEQ and DAX fell more than the S&P 500 (European exposure). These are descriptive — your models test them formally.

B. Irish crisis timeline

Date	Event	Why it matters for you
2003–2007	Credit and property boom (mortgages ≈7×, developer loans ≈11× 1997 levels)	Minsky’s calm boom; your pre-GFC training data
20 Feb 2007	ISEQ peak 9,981 (your data)	Irish decline starts before the global crisis
9 Aug 2007	BNP Paribas freezes three funds (Brunnermeier, p.9–10)	Start of your GFC window

Sep 2007	Northern Rock — first UK bank run in over a century (Brunnermeier)	Contagion to UK/Irish banks
15 Sep 2008	Lehman Brothers fails	Peak global stress
30 Sep 2008	Irish guarantee of six banks' deposits (Acharya et al., p.3)	Bank risk moves to the state
Jan 2009	Anglo Irish Bank nationalised	Domestic banking collapse
9 Mar 2009	ISEQ trough 1,916 (your data)	End of GFC window
23 Apr 2010	Greece requests EU/IMF aid	Start of Irish sovereign-debt window
28 Nov 2010	€85bn EU/IMF programme for Ireland	Sovereign crisis peak for Ireland
26 Jul 2012	Draghi: 'whatever it takes'	End of window; Whelan: turnaround from 2012
15 Dec 2013	Ireland exits the programme	Recovery
23–27 Jun 2016	Brexit vote; ISEQ –7.7% and –9.9%	UK-origin validation event
Feb–Mar 2020	COVID-19; ISEQ –9.9% on 12 Mar	Exogenous crisis window
24 Feb 2022	Russia invades Ukraine; ISEQ –4.6%; ECB hikes from Jul 2022	Energy-shock validation event (euro channel)
Sep 2023 – Jul 2024	CRH, Flutter, Smurfit leave the ISEQ	Sample ends 31 Aug 2023

C. Concept primers with figures from your data

C1. Crisis theory

- **Efficient markets (Fama, 1970).** Returns should be close to unpredictable; volatility is not — see Figure H5. This is why you forecast risk, not price.
- **Minsky (1992).** Hedge → speculative → Ponzi finance; Theorem 1: stable and unstable regimes exist; Theorem 2: long prosperity moves the economy toward the unstable regime; tight money can turn speculative units Ponzi.
- **Kindleberger.** Displacement → credit boom → euphoria → distress → panic → lender of last resort.
- **Reinhart & Rogoff.** Crises recur decades apart, so each generation thinks 'this time is different'.
- **Volatility paradox (Danielsson et al., 2018).** Prolonged below-trend volatility — not high volatility — precedes banking crises; Minsky measured in data (Figure H9).

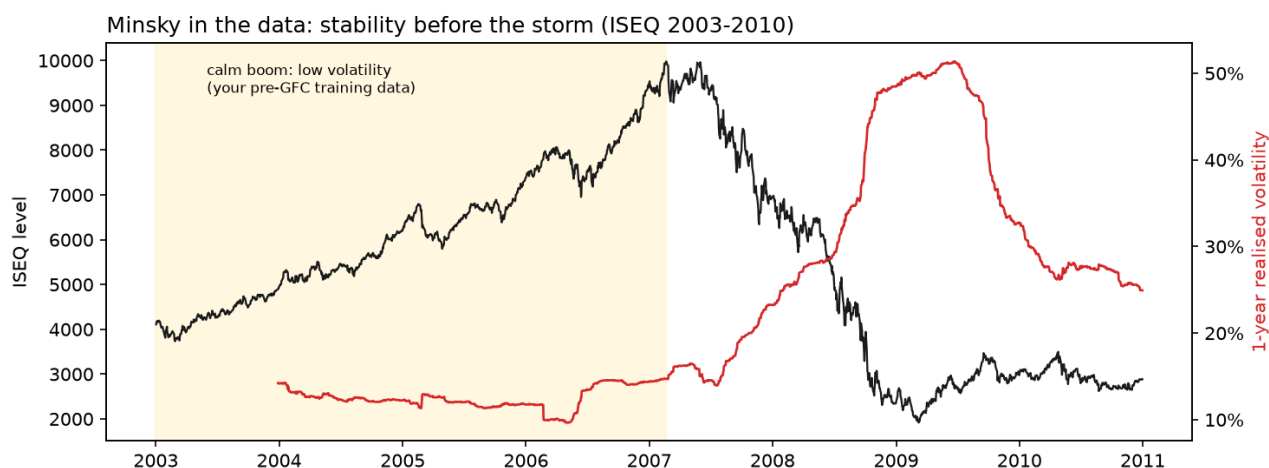


Figure H2. Your data: the ISEQ doubled between 2003 and early 2007 while its 1-year realised volatility stayed around 10–15%; volatility then rose to about 50% in 2008–09. This is what Minsky's 'stability breeds instability' looks like in a market, and it is why RQ3 asks whether a model trained only on the calm boom can cope with the crash.

C2. Contagion and transmission

- **Interdependence vs contagion (Forbes & Rigobon).** Correlations rise mechanically when volatility rises; the adjusted correlation is $\rho^* = \rho / \sqrt{1 + \delta(1 - \rho^2)}$, δ = relative rise in the source market's variance.
- **Heat waves vs meteor showers (Engle, Ito & Lin).** Own-market persistence vs volatility travelling between markets. Your 'domestic' channel (ISEQ) = heat wave; US (S&P 500) and euro (DAX) channels = meteor showers.

- **Non-synchronous trading (Hamao et al.; Forbes & Rigobon).** Markets close at different times. Forbes & Rigobon use two-day average returns; you instead align on the Irish calendar and use foreign closes up to day t (Figure H4).
- **Spillover index (Diebold & Yilmaz).** Share of a market's forecast-error variance caused by other markets; volatility spillovers burst in crises. The 2012 version uses a generalized decomposition that does not depend on variable order and adds directional (to/from) and net spillovers — the method of your exploratory E1 (Figures H7–H8).



Figure H3. Your data: 250-day correlation of daily ISEQ returns with London and Frankfurt (same day) and New York (previous US day). Links with Europe rose towards 0.85 in the Irish debt crisis and COVID — the pattern Forbes & Rigobon warn can reflect volatility rather than contagion. The New York link is weaker and collapses when a few extreme days enter the window (e.g., early 2020), showing how fragile simple correlation is — one reason to use forecasting models and attributions instead.

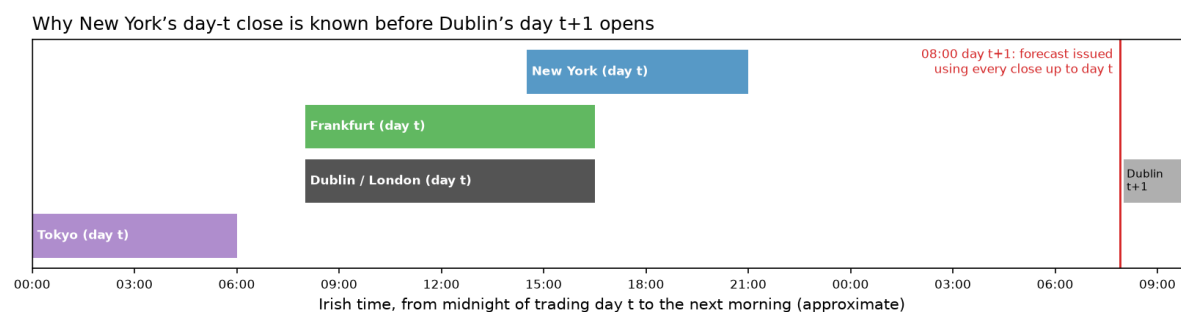


Figure H4. Original diagram: New York closes at about 21:00 Irish time, after Dublin. A forecast issued at 08:00 on day t+1 may legitimately use every market's close up to day t — never day t+1 (Hamao, Masulis & Ng, 1990).

C3. Volatility econometrics

- **Stylised facts (Cont, 2001).** Little autocorrelation in returns, strong and slowly decaying autocorrelation in squared returns (Figure H5).
- **GARCH(1,1).** $\sigma^2(t) = \omega + \alpha \cdot \varepsilon^2(t-1) + \beta \cdot \sigma^2(t-1)$; $\alpha + \beta$ = persistence.
- **HAR (Corsi, 2009).** $RV(t+1) = \beta_0 + \beta_d \cdot \text{daily} + \beta_w \cdot \text{weekly} + \beta_m \cdot \text{monthly}$ (Figure H6).
- **QLIKE (Patton, 2011).** $RV/F - \ln(RV/F) - 1$; robust to noisy volatility proxies; penalises under-prediction more.
- **Diebold–Mariano + HLN.** Test on the average loss difference; Newey–West with $h-1 = 4$ lags for 5-day forecasts; small-sample correction.

ISEQ 2003-2025: returns are close to unpredictable, volatility is persistent

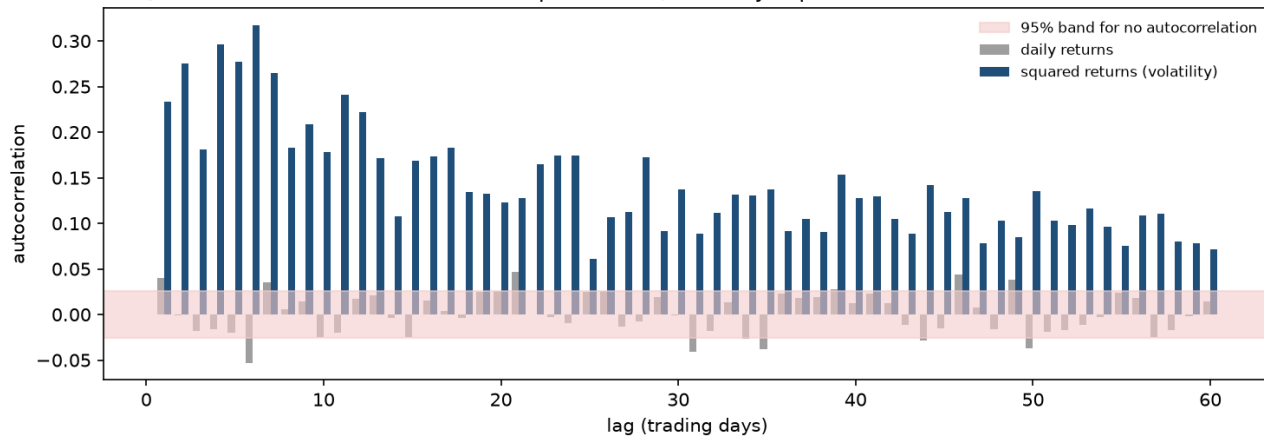


Figure H5. Your data: autocorrelation of ISEQ daily returns (grey) is almost always inside the ‘no autocorrelation’ band, but squared returns (blue) are strongly and persistently autocorrelated even 60 days back. Returns are hard to predict; volatility is not. Long memory like this is what HAR and LSTM are designed to capture.

The three HAR ingredients for the ISEQ during the GFC (2008 - mid-2009)

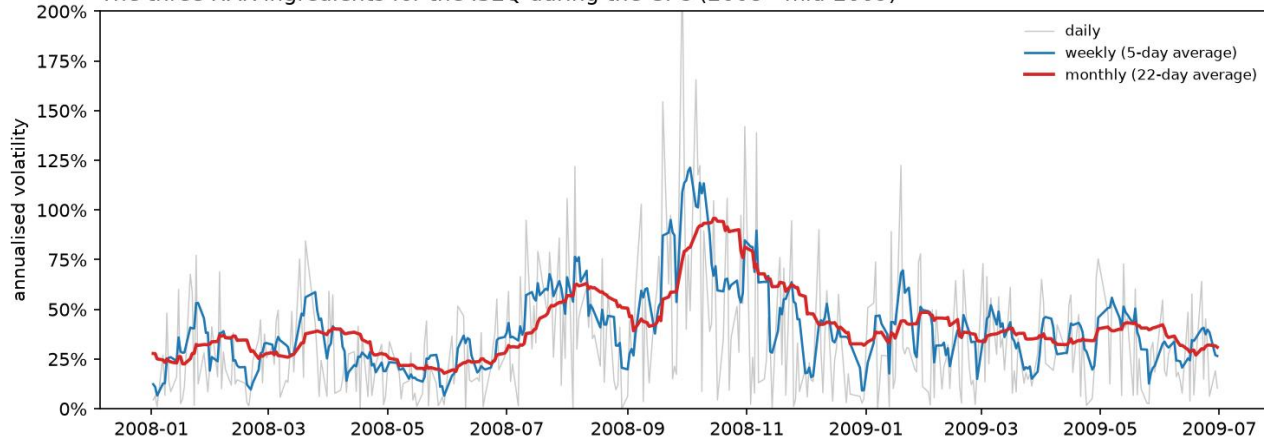


Figure H6. Your data: the three HAR ingredients during the GFC. The daily series (grey) is very noisy; the weekly average (blue) follows turning points; the monthly average (red) carries the slow, persistent level. HAR combines all three; your LSTM sees the same history day by day.

C4. LSTM and C5. Explainable AI (summary)

- **LSTM.** Cell state + gates (input, output; forget gate added by Gers et al., 2000) fix vanishing gradients.
- **SHAP.** Unique additive attribution with local accuracy, missingness, consistency (Lundberg & Lee).
- **Integrated Gradients.** Sensitivity + implementation invariance; completeness; depends on a baseline.
- **Expected gradients / GradientSHAP.** IG averaged over baselines drawn from real data (Erion et al.).
- **Limits.** Correlated features (Aas et al.); mathematical and human-centred problems (Kumar et al.); prefer interpretable models in high-stakes use (Rudin).

C6. Tools of the exploratory extensions (added 30 Sep 2026)

- **Connectedness (E1).** A VAR(4) on log 5-day realised variance of ISEQ, S&P 500 and DAX, re-estimated on each 200-day window. The ISEQ row of the generalized variance decomposition gives own / from-US / from-euro shares — comparable with your SHAP channel shares, but from a linear model of volatility only.
- **One-sided HP trend (E2).** For each year the trend is fitted to past years only; volatility below it counts as ‘calm’. The five-year average of that calm is Danielsson et al.’s crisis indicator.

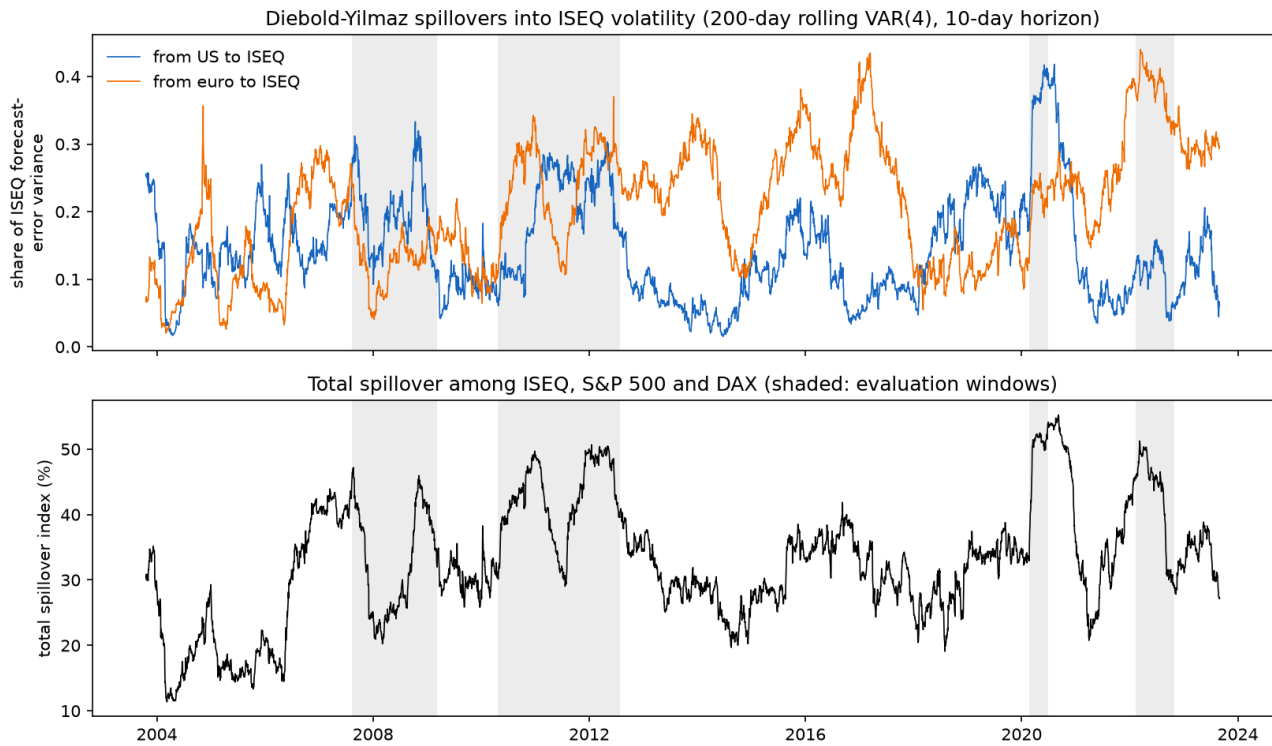


Figure H7. Your data: share of the ISEQ's volatility forecast-error variance coming from the US (blue) and the DAX (orange) in a 200-day rolling VAR, and the total spillover index (bottom). The US share jumps in COVID-19; the DAX share is highest in 2016–17 and 2022; total spillover peaks at 55% in September 2020. Method: Diebold & Yilmaz (2012).

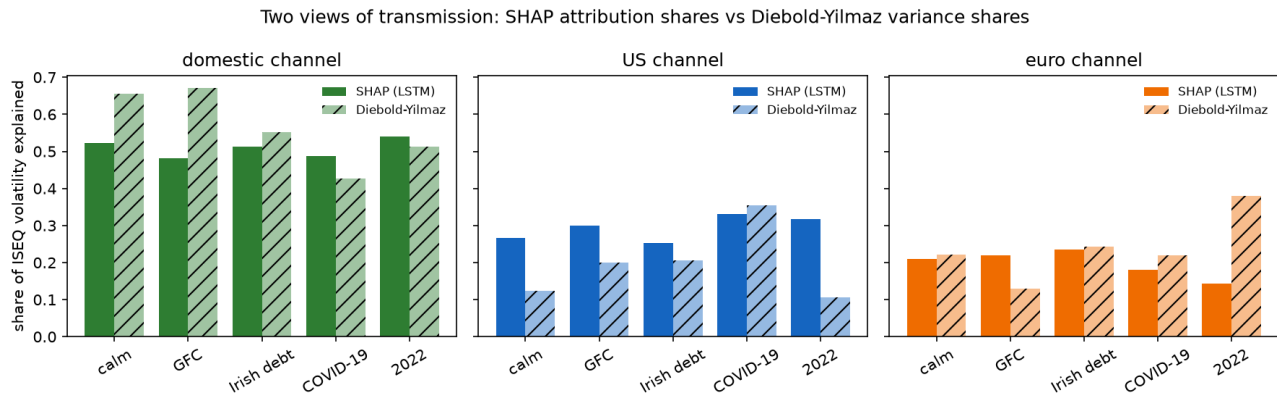


Figure H8. Your data: SHAP channel shares of the LSTM (solid) next to Diebold–Yilmaz shares (hatched) in each window. The two lenses agree that the US share was highest in COVID-19 and differ most in 2022 (Project Handbook, H9).

Volatility paradox check (Danielsson, Valenzuela & Zer, 2018): were crises preceded by calm?

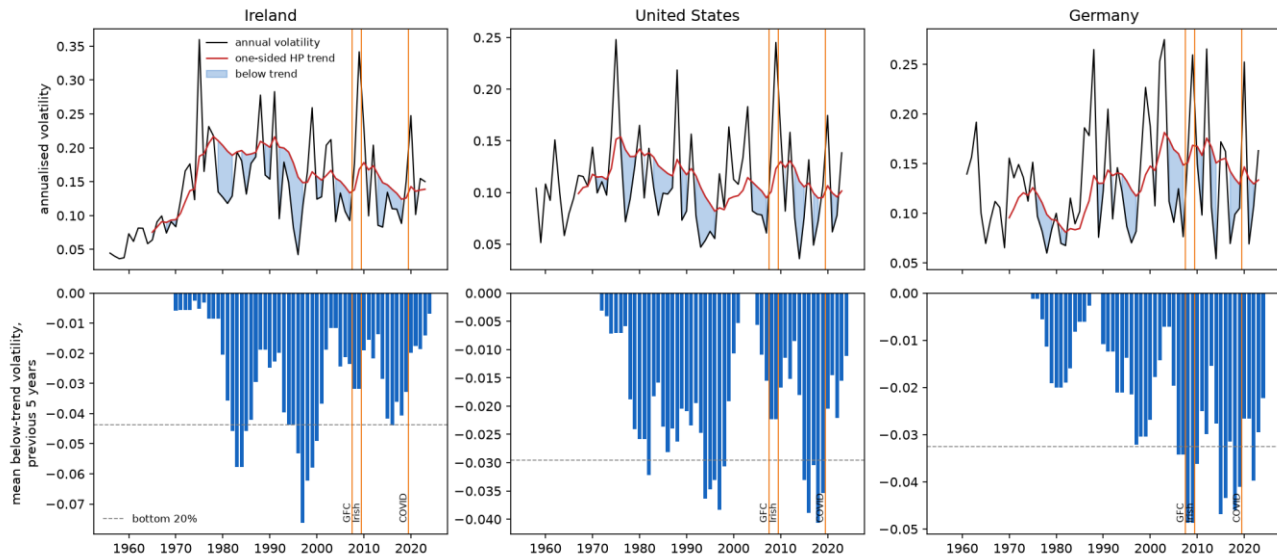


Figure H9. Your data (FRED, from 1955): annual volatility and its one-sided HP trend (top) and the five-year below-trend measure (bottom) for Ireland, the US and Germany. Only Germany was unusually calm before the GFC. Method: Danielsson, Valenzuela & Zer (2018).

C7. Tools of the ML extensions E3–E4 (added 1 Oct 2026)

- **Time-ordered validation.** TimeSeriesSplit always trains on earlier blocks and checks on the next one; a 5-day gap removes the days whose 5-day target overlaps the check block. The class default cv=5 lets a model learn from the future (Bergmeir & Benítez, 2012).
- **The training-range ceiling.** A Random Forest forecast is an average, and an XGBoost forecast a sum, of values learnt from training days; an RBF SVR falls back towards its average on unfamiliar days. None can forecast a crisis bigger than any in its history (Figure H10).
- **Hybrid target (E4).** The learner forecasts only the correction $\log(RV5) - \log(\text{GARCH forecast})$; GARCH carries the level, so the ceiling no longer binds (Figure H11, top). The idea of feeding GARCH into a learner comes from Kim & Won (2018).
- **Loss asymmetry.** QLIKE punishes forecasts that are too low much more than forecasts that are too high; MSE punishes any large error. HAR-X is first on QLIKE and last on MSE (Figure H11, bottom) — always report both.
- **Multiple testing.** With 14 models some will look better by chance. A3 names one primary family and adjusts it with Holm (1979); the other comparisons are descriptive.

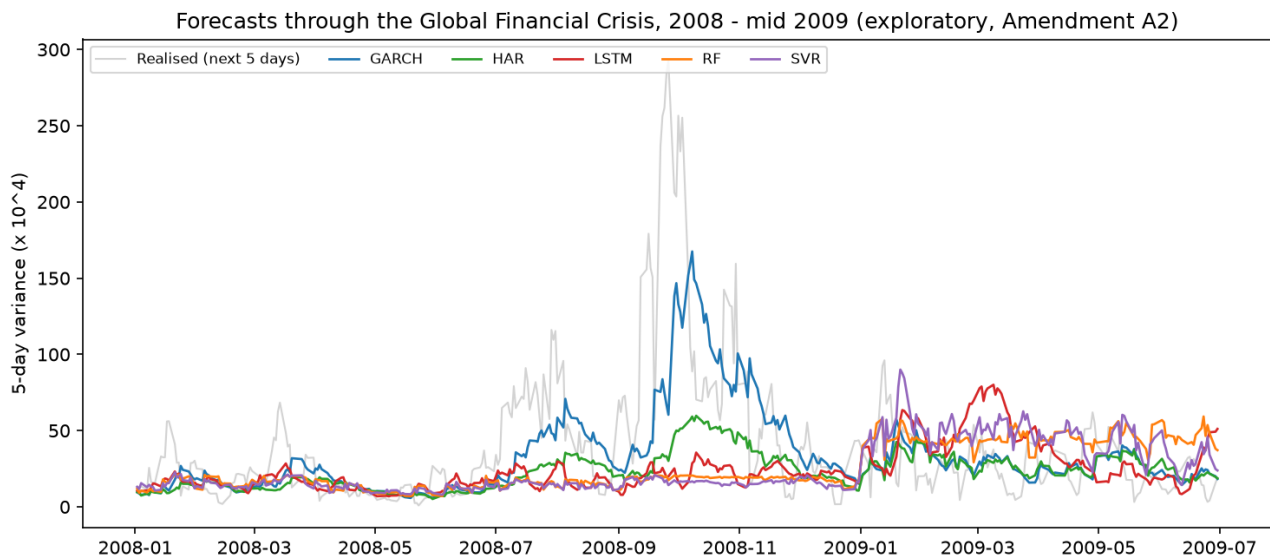


Figure H10. Your data (E3): forecasts through the GFC, 2008 – mid 2009. GARCH climbs with the crisis; the LSTM, Random Forest and SVR stay below $40 (\times 10^{-4})$ while realised variance reaches 292.

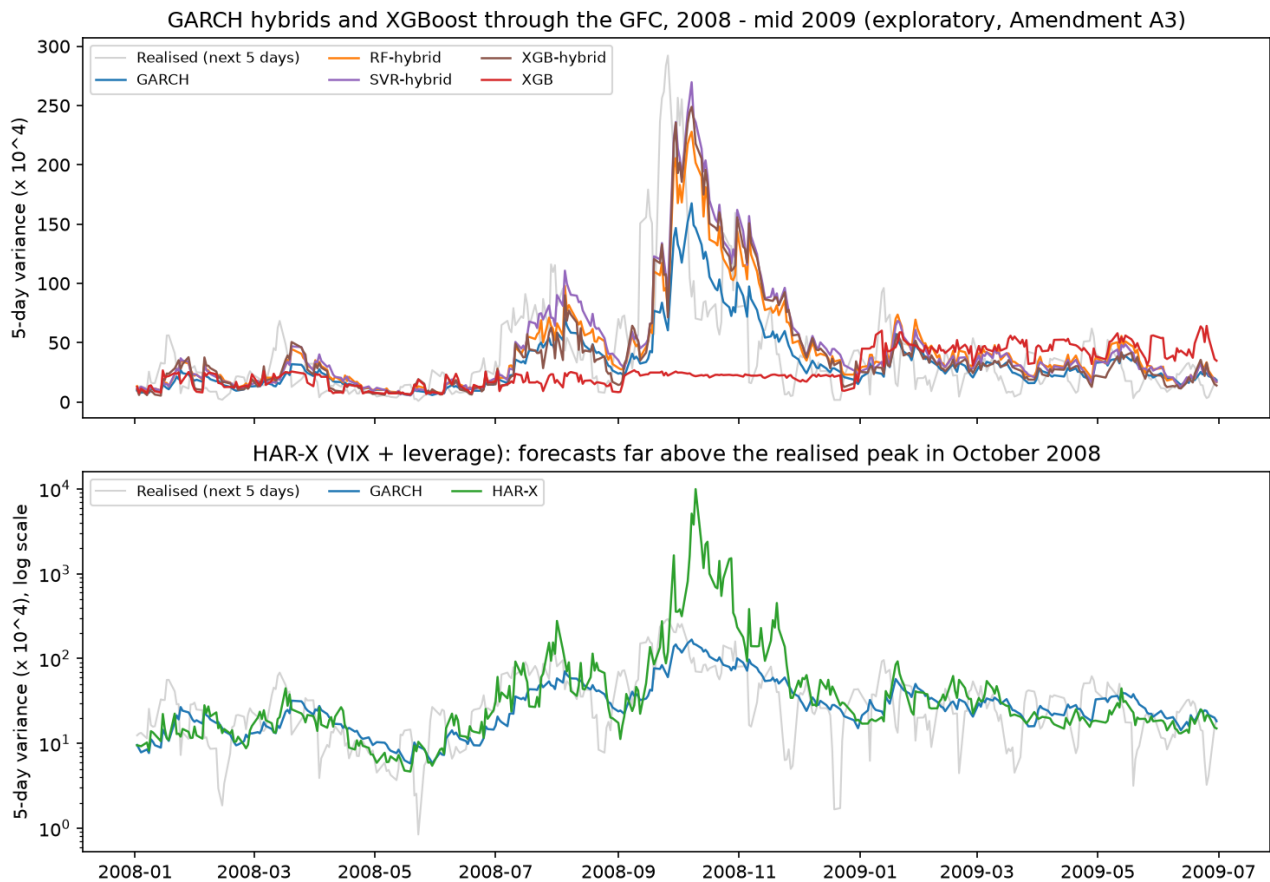


Figure H11. Your data (E4): top, the GARCH hybrids follow the crisis peak (228–270 against 292) while plain XGBoost stays flat; bottom (log scale), HAR-X forecasts reach 9,955 in October 2008.

D. Paper cards

D1. Section 2.2 — Why crises recur

Minsky, H. (1992) The Financial Instability Hypothesis. Levy Institute WP 74.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	A short essay arguing that financial crises come from inside the economy: long periods of stability encourage ever-riskier borrowing until the system becomes fragile and a small shock causes a crash.
Dataset	None — a theoretical essay.
Methodology	Verbal theory building on Keynes; classification of borrowers by how their income covers debt payments.
Results	- Hedge units pay interest and principal from income; speculative units pay interest but must refinance principal; Ponzi units cannot even pay interest. - Theorem 1: an economy has stable and unstable financing regimes. - Theorem 2: long prosperity moves the economy from stable to unstable financing. - The more speculative/Ponzi finance, the more the system amplifies shocks. - Fighting inflation with tight money can turn speculative units into Ponzi units.
How it relates to MY study	The theory behind RQ3: your models learn from the calm 2003–07 boom (Figure H2) and are then tested on the crash. Also frames the 2022 rate shock (tight money).
Must read in the original	Whole paper (10 pages). Key: definitions of the three finance types and the two theorems (pp. 6–8).
Can skim / skip	Nothing — it is short.
Key terms	hedge / speculative / Ponzi finance; financial fragility; deviation-amplifying system
Strengths	Clear, influential, anticipates the GFC mechanism.
Weaknesses / critique	Not formally tested; hard to measure directly.
Use it as · where	Theory support · 2.2; H3; Discussion

Reinhart, C. & Rogoff, K. (2008/2009) This Time Is Different (NBER WP 13882; book 2009).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	A historical database of crises across eight centuries showing that defaults and banking crises recur, that they cluster, and that each generation wrongly believes it is immune.
Dataset	66 countries (Table 1, p.15), all regions including India and China; from the 14th century to 2008.
Methodology	Descriptive statistics on defaults, banking crises, inflation and currency crashes.
Results	- Serial default is nearly universal as countries develop. - Defaults are years or decades apart, creating the ‘this time is different’ illusion. - Crises often start in financial centres and spread through interest-rate and commodity-price shocks. - The US sub-prime crisis was ‘hardly unique’.
How it relates to MY study	Your title; the US-origin expectation for the GFC (H2a); the history lens in your Discussion.
Must read in the original	Abstract (p.2); I. Introduction (pp.3–4); II. First insights (pp.5–15), especially Figure 1 (p.6, share of countries in default).
Can skim / skip	Data appendices (pp.56–125).
Key terms	serial default; ‘this time is different’ syndrome; financial centre
Strengths	Unmatched historical scope.
Weaknesses / critique	Annual macro data; about incidence, not market transmission or forecasting.
Use it as · where	Context + framing · 1, 2.2, 5

Fama, E. (1970) Efficient capital markets. Journal of Finance, 25(2), 383–417.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	The classic review defining weak, semi-strong and strong market efficiency.
Dataset	Review of empirical studies before 1970.
Methodology	Literature review and classification of tests.
Results	- Strong support for weak and semi-strong efficiency; little for the strong form. - Tests of efficiency are joint tests with a model of expected returns.
How it relates to MY study	Why you forecast volatility (predictable) rather than returns (not).
Must read in the original	Introduction and the weak-form evidence section.
Can skim / skip	Detailed older test results.
Key terms	weak / semi-strong / strong form; joint hypothesis
Strengths	Foundational definitions.
Weaknesses / critique	About returns; pre-dates volatility-clustering evidence.
Use it as · where	Context · 2.2; Chapter 3 target choice

Kindleberger, C. & Aliber, R. (2015) Manias, Panics and Crashes, 7th edn.

Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Narrative history showing that manias and crashes follow a recurring pattern.
Dataset	Historical episodes over several centuries.
Methodology	Comparative history built on Minsky.
Results	• Stages: displacement, credit boom, euphoria, distress, panic, lender of last resort. • Crises spread internationally through trade, capital flows and psychology.
How it relates to MY study	Classify the 'displacement' (origin) of each of your three crises.
Must read in the original	The chapter on the anatomy of a typical crisis and the chapter on international contagion (check chapter numbers in your edition).
Can skim / skip	Individual historical cases outside your period.
Key terms	displacement; euphoria; revulsion; lender of last resort
Strengths	Readable, rich history.
Weaknesses / critique	Narrative, selective cases.
Use it as · where	Context · 2.2; origin table in 2.3

Kelly, M. (2009) The Irish credit bubble. UCD CER WP09/32.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Documents the scale of Irish bank lending growth and warns of large bank losses.
Dataset	Irish bank lending and property data to 2008–09.
Methodology	Descriptive analysis and loss scenarios.
Results	• Mortgage lending $\approx 7\times$ its 1997 value by 2008; developer lending $\approx 11\times$.
How it relates to MY study	Irish evidence of Minsky's boom; supports a large domestic channel in the GFC (H2a).
Must read in the original	Whole paper (short working paper): lending figures and conclusions.
Can skim / skip	—
Key terms	credit bubble; developer lending
Strengths	Early, prescient.
Weaknesses / critique	Working paper.
Use it as · where	Support · 2.2–2.3.2; H2a; H3

Cont, R. (2001) Empirical properties of asset returns. Quantitative Finance, 1(2), 223–236.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Lists the statistical regularities common to financial returns.
Dataset	Many assets and markets.
Methodology	Survey.
Results	• Weak return autocorrelation; strong, slowly decaying autocorrelation of absolute/squared returns. • Heavy tails; asymmetry; volatility clustering; leverage effect.
How it relates to MY study	Figure H5 shows these facts in your ISEQ data; justifies log-RV target and Student-t GARCH.
Must read in the original	Section listing the stylised facts.
Can skim / skip	Technical estimation sections.
Key terms	stylised facts; volatility clustering; long memory; leverage effect
Strengths	Clear reference list of facts.
Weaknesses / critique	Descriptive.
Use it as · where	Support · 2.5; Chapter 3

Danielsson, J., Valenzuela, M. & Zer, I. (2018) Learning from history: volatility and financial crises. RFS, 31(7), 2774–2805.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Builds a 211-year, 60-country volatility database and asks whether volatility predicts banking crises. The level does not; long spells of volatility below its trend do — calm encourages credit and leverage ('stability is destabilising').
Dataset	Monthly real stock-index returns, 60 countries, up to 211 years (3,700 country-years; 62 years per country on average); 259 banking crises (Reinhart & Rogoff, 2009); BIS credit data (37 countries, from the 1960s); bank leverage (31 countries, from the 1980s).
Methodology	Annual volatility = standard deviation of 12 monthly returns (July–June); one-sided (recursive) Hodrick–Prescott trend, $\lambda = 5,000$; deviations split into high and low parts; 5-year backward averages; panel logit with decade and region fixed effects; signal-to-noise checks.
Results	• The level of volatility does not predict crises once controls are included. • Prolonged low volatility does: 1% below trend for five years $\rightarrow$ about 1.01% higher crisis probability. • High volatility: weak evidence only. • Low volatility is followed by excess lending and higher bank leverage. • Low volatility works as a crisis indicator with a good signal-to-noise ratio.
How it relates to MY	Empirical anchor for your Minsky framing (RQ3) and the method of exploratory E2. In your data only Germany was

study	unusually calm before the GFC (4th percentile); Ireland was calm but not unusually so (40th).
Must read in the original	Introduction pp.3–6 (findings in brief); Section 2.3 pp.13–14 (the decomposition — your E2 method); Sections 3.1–3.2 pp.19–22 (regression and main result); Conclusion p.32. Pages: LSE accepted version.
Can skim / skip	Control variables (2.5); later robustness tables. The Web Appendix has the same chart for other countries.
Key terms	volatility paradox; one-sided HP filter; high / low volatility; signal-to-noise ratio; excess credit
Strengths	Very long history; simple measure that uses only past data; links calm to credit and leverage.
Weaknesses / critique	Annual volatility from monthly data; banking crises only; the one-sided trend is unstable early in a series; a panel result, not a per-country test.
Use it as · where	Evidence + method · 2.2 (one or two sentences); Chapters 3–4 (E2); Discussion (Minsky)

D2. Section 2.3 — Three crises and Ireland

Brunnermeier, M. (2009) Deciphering the liquidity and credit crunch 2007–2008. JEP, 23(1), 77–100 (NBER WP 14612).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Explains how mortgage losses were amplified into a global crisis, with a day-by-day logbook and four amplification mechanisms.
Dataset	Market indicators 2007–08: ABX mortgage indices, commercial paper, interest-rate spreads, CDS spreads.
Methodology	Economic narrative plus a framework of amplification mechanisms.
Results	• 9 Aug 2007: BNP Paribas froze three funds, unable to value structured products. • Northern Rock suffered the first UK bank run in over a century. • Four mechanisms: balance-sheet loss and margin spirals; lending channel (banks hoard funds); runs on institutions (Bear Stearns, Lehman, Washington Mutual); network effects. • US stock market wealth fell ≈\$8 trillion between Oct 2007 and Oct 2008.
How it relates to MY study	Justifies your GFC start date (9 Aug 2007) and the US-origin expectation (H2a: US channel = S&P 500).
Must read in the original	Banking trends (pp.4–8); event logbook (pp.9–21; Figures 1–4 on pp.10, 12, 14, 20); amplifying mechanisms (pp.22–29; spirals figure p.24); conclusion (p.30).
Can skim / skip	Detailed product descriptions if short of time.
Key terms	originate-and-distribute; loss spiral; margin spiral; liquidity hoarding; bank run
Strengths	Clear mechanism; primary-event chronology.
Weaknesses / critique	Written mid-crisis; US-centred.
Use it as · where	Context + support · 2.3.2; H2a

Bekaert, G., Ehrmann, M., Fratzscher, M. & Mehl, A. (2014) The global crisis and equity market contagion. JF, 69(6), 2597–2649 (ECB WP 1381).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Asks how the GFC spread across world equity markets; finds that domestic weaknesses mattered more than direct links to the US.
Dataset	415 country-sector equity portfolios in 55 markets; weekly data, ≈725 observations each.
Methodology	Factor asset-pricing model (US, global-financial and domestic factors); contagion = unexplained rise in factor loadings and residual correlations during the crisis.
Results	• Little systematic contagion from US markets despite the US origin. • Systematic contagion from domestic markets to domestic portfolios. • Weak fundamentals, poor policies and weak institutions → more contagion and larger falls. • Debt and deposit guarantees partly shielded domestic equities (wake-up call hypothesis).
How it relates to MY study	Key basis for H2a: expect a large domestic channel for Ireland in the GFC, not only US channels. Matches the ISEQ's +7.9% on guarantee day (Figure H1).
Must read in the original	Introduction (pp.8–10); 2.1.1 interdependence vs contagion (p.12); 3.2 contagion (pp.20–22); 3.3 channels (pp.23–24); conclusions (pp.25–26); Figure 3 (p.21).
Can skim / skip	Estimation diagnostics (pp.15–17) and appendix.
Key terms	interdependence; contagion; factor loadings; wake-up call hypothesis
Strengths	Very large cross-section; clear definition.
Weaknesses / critique	Result depends on the factor model; not a forecasting study.
Use it as · where	Central support · 2.3.2, 2.4; H2a

Acharya, V., Drechsler, I. & Schnabl, P. (2014) A Pyrrhic victory? JF, 69(6), 2689–2739.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Shows that bank bailouts transferred risk to governments, creating a two-way loop between bank and sovereign credit risk — with Ireland as the opening case.
Dataset	CDS spreads for European banks and sovereigns (2007 onwards); bond yields from Datastream.
Methodology	Theoretical model of the loop; event study around bailouts; panel regressions.
Results	• Ireland guaranteed six banks' deposits on 30 Sep 2008. • Irish bank CDS fell overnight from ≈400 to ≈150 bp while sovereign CDS rose sharply. • By autumn 2010, the Irish–German spread widened and Ireland needed a bailout. • Across Europe, bank/sovereign CDS: 63/14 → 148/48 → 184/112 bp (before/during/after bailouts).

	• After bailouts, sovereign CDS changes help explain bank CDS changes.
How it relates to MY study	Explains how your GFC window led into your Irish debt-crisis window; basis for a strong domestic channel in 2010–12 (H2b).
Must read in the original	Introduction and Figure 1 (pp.2–4, the Irish case); empirical analysis: data (p.23), sovereign risk trigger (pp.24–27, Figures 8–9 p.25), feedback loop (pp.28–35, Table III p.30); conclusion (p.40).
Can skim / skip	Model sections (pp.8–22) unless interested in the theory.
Key terms	bank–sovereign loop; CDS; bailout; guarantee
Strengths	Clear mechanism; clean natural experiment.
Weaknesses / critique	Credit markets, not equity volatility.
Use it as · where	Support · 2.3.2–2.3.3; H2b

Honohan (2010); Regling & Watson (2010); Nyberg (2011) — official Irish reports.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	The three official investigations into why Irish banks failed.
Dataset	Bank, regulator and government documents; interviews.
Methodology	Official inquiries.
Results	• Property-lending concentration and foreign wholesale funding. • Light-touch regulation; failed macro-prudential policy (Honohan). • Largely home-grown crisis, triggered by the global crisis (Regling & Watson). • Herding by banks and authorities (Nyberg).
How it relates to MY study	Primary historical evidence for the domestic dimension of the GFC in Ireland (H2a).
Must read in the original	Executive summaries and conclusions of each report.
Can skim / skip	Detailed institutional chapters.
Key terms	macro-prudential policy; herding; wholesale funding
Strengths	Authoritative.
Weaknesses / critique	Not academic tests; confirm wording and pages.
Use it as · where	Context + support · 2.3.2; Discussion

Lane, P. (2012) The European sovereign debt crisis. JEP, 26(3), 49–68.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Explains the euro-area debt crisis as the result of a flawed monetary-union design.
Dataset	Euro-area macro and fiscal data.
Methodology	Economic analysis.
Results	• Flawed euro design; poor understanding of monetary-union fragility. • No banking union or euro-level buffers; crisis-management errors.
How it relates to MY study	Euro-area origin of your 2010–12 window (H2b).
Must read in the original	Sections on origins, the Irish case and policy responses (free on aeaweb.org).
Can skim / skip	—
Key terms	monetary union; banking union; sovereign spreads
Strengths	Clear, by a later Central Bank of Ireland Governor.
Weaknesses / critique	Mid-crisis perspective.
Use it as · where	Context · 2.3.3; H2b

Whelan, K. (2014) Ireland's economic crisis. J. Macroeconomics, 39(B), 424–440.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Reviews Ireland's boom, crash and recovery.
Dataset	Irish macro data, 2000s–2013.
Methodology	Economic review.
Results	• Mid-2000s demand above potential due to a property bubble. • Better fiscal policy or regulation could have prevented the crisis. • Crisis management and the turnaround from 2012 were reasonably successful.
How it relates to MY study	Irish narrative; supports ending your window in 2012.
Must read in the original	Whole paper (UCD WP 13/06 version).
Can skim / skip	—
Key terms	output gap; fiscal consolidation
Strengths	Concise overview.
Weaknesses / critique	Little on markets.
Use it as · where	Context · 2.3.3

Corbet, S. & Twomey, C. (2015) European equity market contagion: Ireland's sovereign debt crisis. EFAJ, 2015(3), 15–34.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Tests whether the collapse of Irish equities spilled over to European markets.
Dataset	Daily European equity returns; 2007–10 crisis and post-troika period.
Methodology	DCC multivariate GARCH.
Results	• Strong Irish–European equity contagion 2007–10. • Contagion eased dramatically after troika intervention.
How it relates to MY study	The Irish study closest to yours, by Dr Twomey. Test: does your euro (DAX) share fall after Nov 2010?
Must read in the original	Whole paper (≈20 pages) — essential before contacting Dr Twomey.
Can skim / skip	—
Key terms	DCC-GARCH; troika; contagion
Strengths	Irish focus; policy result.
Weaknesses / critique	Correlation-based; no channels by origin; no forecasting.
Use it as · where	Central support + contrast · 2.3.3, 2.4; H2b

Baker, S. et al. (2020) The unprecedented stock market reaction to COVID-19. RAPS, 10(4), 742–758 (NBER WP 26945).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Shows that no earlier pandemic moved stock markets like COVID-19 and explains why.
Dataset	US daily market moves since 1900; volatility since 1985; next-day newspaper explanations.
Methodology	Text-based classification of the reported causes of large daily moves.
Results	• >1,100 US daily moves >2.5% (1900–2019); none attributed to pandemics. • ≈ two dozen COVID-attributed jumps, 24 Feb – 20 Apr 2020. • COVID excess mortality ≈1/25th of the Spanish Flu's. • Best explanation: restrictions and social distancing in a service economy.
How it relates to MY study	COVID was genuinely new → a model trained on past crises has never seen it (H3); broad simultaneous shock (H2c).
Must read in the original	Main text pp.3–15 (jump characterisation p.4; explanations p.8; conclusions p.13); Table 1 (p.20); Figures 1–3 (pp.19–22).
Can skim / skip	References.
Key terms	market jumps; equity market volatility tracker; non-pharmaceutical interventions
Strengths	Long history; transparent method.
Weaknesses / critique	US only; early 2020.
Use it as · where	Support · 2.3.4; H2c; H3; Discussion

Akhtaruzzaman, M., Boubaker, S. & Sensoy, A. (2021) Financial contagion during COVID-19 crisis. FRL, 38, 101604.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Measures how COVID contagion spread between China and G7 markets through financial and non-financial firms.
Dataset	Daily stock returns of financial and non-financial firms, China and G7; pre-COVID 1 Jan 2013 – 30 Dec 2019; COVID 31 Dec 2019 – 20 Mar 2020.
Methodology	VARMA(1,1)-DCC-GARCH correlations; optimal hedge ratios; directional spillover index.
Results	• Correlations rose significantly for all firm types, much more for financial firms. • China and Japan were net transmitters; hedging costs rose.
How it relates to MY study	Supports H2c (broad, simultaneous rise) and financial firms as transmitters.
Must read in the original	3.1 Data (p.4); 4.1 DCCs (p.10, Figure 3 p.7); 4.3 spillovers (pp.13–16, Figures 6–7); conclusion (p.16).
Can skim / skip	Portfolio weights section (4.2) if short of time.
Key terms	DCC; net transmitter; hedge ratio
Strengths	Firm-level detail.
Weaknesses / critique	Only ≈2.5 months of COVID data; no Ireland.
Use it as · where	Support · 2.3.4; H2c

Li, H., Ahmed, S. & Chevapatrakul, T. (2016) Volatility spillovers around the Brexit referendum. Nottingham CFCM WP 16/06.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Measures intraday volatility spillovers among four big European markets around the Brexit vote.
Dataset	15-minute intraday data; France, Germany, Switzerland, UK; 4 Jan – 30 Sep 2016.
Methodology	Diebold–Yilmaz total, directional, net and pairwise spillovers; full and rolling samples.
Results	• Strong interdependence among the four markets (full-sample spillover table: Table 3, p.13). • Overall: France and Germany net transmitters; UK and Switzerland net receivers. • Around referendum day: France and the UK net transmitters.
How it relates to MY study	Competing evidence for H2d (tested in your FTSE-for-DAX robustness run) — the UK was a transmitter only around the vote. Compare with the euro share in Brexit check too.

Must read in the original	Introduction (pp.3–5); 3.1 data (p.9); 3.2 full sample (pp.12–13); 3.3.3–3.3.4 net spillovers (pp.16–20); conclusion (p.21).
Can skim / skip	Econometric framework details (pp.6–8).
Key terms	net transmitter / receiver; pairwise spillover
Strengths	High-frequency data.
Weaknesses / critique	Four large markets; no Ireland.
Use it as · where	Contrast · 2.3.5; H2d

Li, H. (2020) Volatility spillovers under the uncertainty of Brexit. Economic Modelling, 84, 1–12.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Journal follow-up on Brexit volatility spillovers in Europe.
Dataset	European stock markets, 2016 onwards (list to confirm).
Methodology	Time-varying multivariate GARCH; variance decomposition; volatility impulse responses.
Results	• UK influence decreased from the start of the campaign. • The vote raised volatility; its effect on co-volatility persisted.
How it relates to MY study	Second competing source on H2d.
Must read in the original	Abstract, data section, conclusion.
Can skim / skip	—
Key terms	volatility impulse response
Strengths	Peer-reviewed.
Weaknesses / critique	Check whether Ireland is included.
Use it as · where	Contrast · 2.3.5; H2d

Bougou, W. & Yatié, A. (2022) The impact of the Ukraine–Russia war on world stock market returns. Economics Letters, 215, 110516.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Short paper measuring the war's impact on world stock returns.
Dataset	Daily returns, 94 countries, 22 Jan – 24 Mar 2022.
Methodology	Panel regressions around the invasion.
Results	• Negative effect on returns, strongest in the first two weeks after 24 Feb.
How it relates to MY study	Background for the 2022 validation event (H2e: euro channel).
Must read in the original	Whole paper (a short letter).
Can skim / skip	—
Key terms	event window
Strengths	Wide coverage.
Weaknesses / critique	Returns only; short window.
Use it as · where	Context · 2.3.5; H2e

Kearney, C. (1998) Causes of volatility in a small, internationally integrated stock market: Ireland, 1975–1994. J. Financial Research, 21(1).	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Early study of what drives Irish stock market volatility.
Dataset	Irish market, 1975–1994.
Methodology	Read the paper.
Results	• Treats Ireland as small and internationally integrated — read for findings.
How it relates to MY study	Historical precedent that foreign factors matter for Irish volatility.
Must read in the original	Introduction, results and conclusion.
Can skim / skip	—
Key terms	small open economy
Strengths	Irish-specific.
Weaknesses / critique	Old, pre-euro sample.
Use it as · where	Context · 2.3.1; H2d

Gallagher, L. & Twomey, C. (1998) Identifying the source of mean and volatility spillovers in Irish equities: a multivariate GARCH analysis. The Economic and Social Review, 29(4), 341–356.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Finds UK spillovers into individual Irish stocks.
Dataset	Ten largest Irish stocks; Irish and UK markets.

Methodology	Multivariate VAR-GARCH.
Results	• Significant mean and volatility spillovers from the UK to Irish stocks.
How it relates to MY study	Why the FTSE robustness run matters (H2d).
Must read in the original	Whole paper.
Can skim / skip	—
Key terms	mean spillover; volatility spillover
Strengths	Irish; multivariate GARCH.
Weaknesses / critique	1990s, pre-euro.
Use it as · where	Support · 2.3.1; H2d

D3. Section 2.4 — How shocks travel

Forbes, K. & Rigobon, R. (2002) No contagion, only interdependence. JF, 57(5), 2223–2261 (NBER WP 7267).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Shows that the standard test for contagion is biased because correlations rise automatically with volatility; after correcting, famous ‘contagions’ disappear.
Dataset	Stock markets in the 1997 East Asian crisis, 1994 Mexican crisis and 1987 US crash.
Methodology	Correlation tests with and without a volatility adjustment; two-day rolling-average returns to handle different opening hours; interest-rate controls.
Results	• Unadjusted correlation is biased upward in turmoil. • Unadjusted tests find contagion in several countries; adjusted tests find virtually none. • Crisis co-movement reflects strong existing links (interdependence). • The adjusted measure can still be biased by endogeneity or common shocks.
How it relates to MY study	Your key caution: a rising SHAP share in a crisis may reflect volatility, not a new channel (Figure H3). Their two-day averaging is an alternative to your time alignment.
Must read in the original	Introduction (pp.3–4); 2.2 framework (pp.10–11); 3.1 bias in the correlation coefficient (pp.13–16) — the core; Tables 2–3 (pp.22–24); conclusions (pp.36–37).
Can skim / skip	Robustness tests 4.1 (pp.25–30).
Key terms	interdependence; contagion; conditional correlation; heteroskedasticity bias
Strengths	Simple, powerful critique.
Weaknesses / critique	Assumes no endogeneity/common shocks.
Use it as · where	Method + caution · 2.4; Limitations

Engle, R., Ito, T. & Lin, W.-L. (1990) Meteor showers or heat waves? Econometrica, 58(3), 525–542 (NBER WP 2609).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Tests whether volatility stays local (heat wave) or travels between markets (meteor shower).
Dataset	Yen/dollar exchange rate across Tokyo, European and New York segments.
Methodology	ARCH models across intra-daily segments; volatility VAR.
Results	• The heat-wave hypothesis is rejected: volatility spills over between markets. • News in one market affects volatility in the next.
How it relates to MY study	Ready-made framing for your domestic vs foreign channels.
Must read in the original	Abstract and introduction; heat-wave section 4.1 (p.11); tables of tests (pp.19–20).
Can skim / skip	Estimation details.
Key terms	heat wave; meteor shower; market dexterity
Strengths	Elegant hypothesis.
Weaknesses / critique	FX, 1980s; scanned PDF is hard to read — use JSTOR version.
Use it as · where	Framing · 2.4; RQ2

Hamao, Y., Masulis, R. & Ng, V. (1990) Correlations in price changes and volatility across international stock markets. RFS, 3(2), 281–307.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Studies spillovers between Tokyo, London and New York using opening and closing prices.
Dataset	Daily open and close prices of major indices.
Methodology	ARCH-family models.
Results	• Volatility spillovers from New York to Tokyo, London to Tokyo, New York to London.
How it relates to MY study	Justifies your time-zone alignment (Figure H4).
Must read in the original	Introduction and conclusion.
Can skim / skip	—
Key terms	open-to-close vs close-to-open returns

Strengths	Classic.
Weaknesses / critique	Three markets.
Use it as - where	Method - 2.4; Chapter 3

King, M. & Wadhwani, S. (1990) Transmission of volatility between stock markets. RFS, 3(1), 5–33.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Explains rising cross-market correlation around the 1987 crash through traders inferring information from other markets.
Dataset	New York, London, Tokyo, 1987.
Methodology	Contagion model; hourly data.
Results	• Mistakes in one market spread to others; correlation rises with volatility.
How it relates to MY study	Early theory of contagion as information spillover.
Must read in the original	Introduction and model intuition.
Can skim / skip	Technical appendix.
Key terms	information contagion
Strengths	Theory + data.
Weaknesses / critique	Single event.
Use it as - where	Context - 2.4

Kearney, C. (2000) Determination and international transmission of stock market volatility. Global Finance Journal, 11, 31–52.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Low-frequency study of what drives volatility and how it moves between major markets.
Dataset	Monthly 1973–1994: UK, France, Germany, Japan, US.
Methodology	Efficient portfolios; cointegration; conditional volatility transmission.
Results	• Cointegrating links between world, European and Japanese/US portfolios; volatility transmission.
How it relates to MY study	Transmission evidence by an Irish-based author.
Must read in the original	Introduction and conclusions.
Can skim / skip	—
Key terms	cointegration
Strengths	Macro determinants.
Weaknesses / critique	Monthly, pre-2000.
Use it as - where	Context - 2.4

Diebold, F. & Yilmaz, K. (2009) Measuring return and volatility spillovers. Economic Journal, 119(534), 158–171 (NBER WP 13811).	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Introduces the spillover index and shows volatility spillovers burst during crises.
Dataset	19 global equity markets, weekly, Jan 1992 – Nov 2007; weekly range-based (Garman–Klass) volatility.
Methodology	VAR(2) forecast-error variance decomposition (10-week horizon); 200-week rolling windows.
Results	• Return spillovers: gentle upward trend, no bursts. • Volatility spillovers: no trend, clear bursts in crises.
How it relates to MY study	Evidence that foreign influence bursts in crises (H2); your PhD validation tool. Note they used Garman–Klass — you rejected it for Yahoo ISEQ data (stale opens): a good methodological point.
Must read in the original	2 Spillover index (pp.3–5) — core; 4 spillover tables (pp.6–7, Tables 3–4 pp.14–15); 5 rolling plots (p.8, Figures pp.16–19); 7 summary (p.10).
Can skim / skip	6 robustness.
Key terms	forecast-error variance decomposition; spillover index; Cholesky ordering
Strengths	Simple, widely used.
Weaknesses / critique	Depends on VAR ordering (fixed in their 2012 version).
Use it as - where	Support + future method - 2.4; Future research

Diebold, F. & Yilmaz, K. (2012) Better to give than to receive: predictive directional measurement of volatility spillovers. IJF, 28(1), 57–66.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Improves the 2009 spillover index with a generalized VAR decomposition that does not depend on variable ordering, and adds directional (to/from), net and pairwise spillovers; applies them to US asset markets.
Dataset	Daily range-based volatility of the S&P 500, the 10-year Treasury yield, US dollar index futures and the DJ-UBS commodity index; 25 Jan 1999 – 29 Jan 2010 (2,771 days).
Methodology	VAR(4); generalized forecast-error variance decomposition (Pesaran & Shin, 1998) at a 10-day horizon, rows normalised; full-sample spillover table plus 200-day rolling windows; sensitivity to VAR order (2–6) and horizon (5–10 days).

Results	• Full-sample total spillover 12.6% — cross-market spillovers were limited. • Spillovers stayed limited until the crisis that began in 2007, then rose. • After Lehman (September 2008) large spillovers ran from the stock market to the other markets. • The index is robust to the VAR order and horizon.
How it relates to MY study	The method of your exploratory E1 (VAR(4), 10-day horizon, 200-day windows) applied to ISEQ, S&P 500 and DAX volatility and compared with SHAP. Your total spillover is 34.0%; the ISEQ is a net receiver (−8.4).
Must read in the original	Section 2 pp.58–59 (variance shares; total, directional and net spillovers — the formulas); Sections 3.1–3.3 pp.59–61 (data, spillover table, rolling index). Journal page numbers.
Can skim / skip	Net pairwise spillover figures (pp.62–65); FX and commodity detail.
Key terms	generalized variance decomposition; total / directional / net spillover; spillover table; rolling window
Strengths	Order-invariant; simple; the standard connectedness tool.
Weaknesses / critique	Linear VAR; rolling-window choices matter; measures co-movement of volatility, not causation.
Use it as · where	Method + context · 2.4; Chapter 3 and 4 (E1)

Pesaran, M.H. & Shin, Y. (1998) Generalized impulse response analysis in linear multivariate models. Economics Letters, 58(1), 17–29.

Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Proposes impulse responses (and variance decompositions) that do not depend on the order of the variables in a VAR, using the historical correlation of shocks instead of an orthogonalisation.
Dataset	None — a methodological note.
Methodology	Generalized impulse responses for VAR models.
Results	• Responses do not depend on variable ordering. • They equal the Cholesky responses when the shocked variable is ordered first. • Generalized variance shares need not add to one — hence Diebold & Yilmaz normalise each row.
How it relates to MY study	Why your E1 shares do not depend on listing ISEQ, S&P 500 and DAX in a particular order (your code was checked for this).
Must read in the original	The definitions of generalized impulse responses and variance decompositions (short paper).
Can skim / skip	Extensions to cointegrated models.
Key terms	generalized impulse response; orthogonalised (Cholesky) decomposition; variable ordering
Strengths	Removes an arbitrary modelling choice.
Weaknesses / critique	Shares must be normalised; correlated shocks are shared, so interpretation stays descriptive.
Use it as · where	Method · Chapter 3 (E1)

Sahiner, M. (2023) Volatility spillovers and contagion during major crises: a deep learning early warning approach. Computational Economics.

Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Combines spillover econometrics with deep learning across major crashes.
Dataset	International stock markets, 3 Jul 1997 – 9 Mar 2021.
Methodology	DCC, GARCH-BEKK, Diebold–Yilmaz; deep-learning early warning.
Results	• External shocks have long-lasting domestic effects in crises. • Asian stress transmitted mainly via Hong Kong.
How it relates to MY study	Closest prior work; your difference: SHAP channel attribution, Ireland, cross-crisis test.
Must read in the original	Introduction, method overview, crisis results, conclusion.
Can skim / skip	—
Key terms	early warning system; BEKK
Strengths	Long sample.
Weaknesses / critique	No explanation of channels; no Ireland.
Use it as · where	Contrast · 2.4; 2.7 gap

Pericoli & Sbracia (2003); Dungey et al. (2005); Bekaert, Harvey & Ng (2005).

Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Definitions and methods for measuring contagion.
Dataset	Reviews; factor-model study.
Methodology	Surveys; factor model.
Results	• Several competing definitions of contagion. • Many tests fit a common latent-factor framework. • Contagion as excess correlation of factor-model residuals.
How it relates to MY study	Definitions for 2.4.
Must read in the original	Definition sections.
Can skim / skip	Technical sections.

Key terms	latent factor; residual correlation
Strengths	Organise the field.
Weaknesses / critique	Confirm details.
Use it as · where	Context · 2.4

D4. Section 2.5 — Forecasting volatility

Engle (1982); Bollerslev (1986).	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	ARCH and GARCH: the foundations of volatility modelling.
Dataset	UK inflation (Engle); general.
Methodology	Conditional variance models.
Results	• Variance depends on past shocks (ARCH) and past variance (GARCH).
How it relates to MY study	Your first benchmark.
Must read in the original	Model definitions.
Can skim / skip	Proofs.
Key terms	conditional heteroskedasticity; persistence
Strengths	Foundational.
Weaknesses / critique	Symmetric response (GJR adds leverage).
Use it as · where	Method · 2.5; Chapter 3

Hansen, P. & Lunde, A. (2005) Does anything beat a GARCH(1,1)? JAE, 20(7), 873–889.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Compares 330 ARCH-type models.
Dataset	DM–USD exchange rate; IBM returns.
Methodology	Realised-variance evaluation; superior predictive ability test.
Results	• Exchange rate: GARCH(1,1) not beaten. • IBM: models with a leverage effect do better.
How it relates to MY study	Sets the bar for your LSTM; suggests GJR-GARCH as a robustness check.
Must read in the original	Introduction, results summary, conclusion.
Can skim / skip	Full model list.
Key terms	SPA test; leverage effect
Strengths	Exhaustive.
Weaknesses / critique	Two series.
Use it as · where	Contrast · 2.5; H1

Andersen & Bollerslev (1998); Andersen, Bollerslev, Diebold & Labys (2003).	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Realised volatility: better yardsticks and better models.
Dataset	High-frequency FX data.
Methodology	Realised volatility; long-memory models of log RV.
Results	• GARCH looks poor against noisy squared returns but good against realised volatility. • Modelling log RV directly forecasts well.
How it relates to MY study	Why 5-day RV and log target.
Must read in the original	Introductions and RV definitions.
Can skim / skip	Technical proofs.
Key terms	realised volatility; measurement noise
Strengths	Foundational.
Weaknesses / critique	Intraday data.
Use it as · where	Method · 2.5; Chapter 3

Corsi, F. (2009) A simple approximate long-memory model of realized volatility. JFEC, 7(2), 174–196.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	The HAR model: three volatility averages in one regression.
Dataset	High-frequency financial data.
Methodology	OLS on daily, weekly, monthly RV.
Results	• Reproduces long memory and fat tails; strong forecasts.
How it relates to MY	Your second benchmark (Figure H6).

study	
Must read in the original	Model section and forecasting results.
Can skim / skip	—
Key terms	heterogeneous market hypothesis; long memory
Strengths	Simple, strong.
Weaknesses / critique	Linear.
Use it as · where	Method · 2.5; H1

Patton, A. (2011) Volatility forecast comparison using imperfect volatility proxies. J. Econometrics, 160(1), 246–256.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Shows which loss functions rank volatility forecasts correctly when true volatility is unobserved.
Dataset	Theory; IBM application.
Methodology	Analytical derivations.
Results	- Noisy proxies can make worse forecasts look better under some losses (affects test size). - Less noisy proxies reduce but do not remove distortion. - Robust family nests MSE and QLIKE.
How it relates to MY study	Justifies MSE + QLIKE (QLIKE primary).
Must read in the original	1 Introduction (pp.1–2); 2.1 squared returns as proxy (p.3); 3 robust loss class (pp.5–6); Figure 2 (p.7); conclusion (p.8).
Can skim / skip	Proofs.
Key terms	volatility proxy; robust loss; QLIKE
Strengths	Rigorous, practical.
Weaknesses / critique	Theory-heavy.
Use it as · where	Method · 2.5; Chapter 3

Diebold & Mariano (1995); Harvey, Leybourne & Newbold (1997).	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Formal tests of whether one forecast beats another.
Dataset	General.
Methodology	Test on mean loss differential; small-sample correction.
Results	DM is robust to autocorrelation; HLN corrects small samples and multi-step forecasts.
How it relates to MY study	Your RQ1 test.
Must read in the original	Test statistic sections.
Can skim / skip	—
Key terms	loss differential; Newey–West; HLN
Strengths	Standard.
Weaknesses / critique	Nested-model issues.
Use it as · where	Method · 2.5; Chapter 3

Hochreiter, S. & Schmidhuber, J. (1997) Long short-term memory. Neural Computation, 9(8), 1735–1780.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Introduces the LSTM to learn long time lags.
Dataset	Artificial benchmark tasks.
Methodology	New recurrent architecture with a constant error carousel.
Results	Solves tasks with very long time lags where plain RNNs fail.
How it relates to MY study	Your model; cite in 2.5 and 3.
Must read in the original	Introduction and the architecture section (first ≈9 pages).
Can skim / skip	Experiments (pp.9–20).
Key terms	constant error carousel; gates
Strengths	Foundational.
Weaknesses / critique	Original had no forget gate (Gers et al., 2000).
Use it as · where	Method · 2.5; Chapter 3

Fischer & Krauss (2018) EJOR 270(2); Kim & Won (2018) ESWA 103.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	LSTM for stock direction (S&P 500 constituents) and LSTM–GARCH hybrids for KOSPI 200 volatility.
Dataset	S&P 500 constituents 1992–2015; KOSPI 200.
Methodology	LSTM vs RF/DNN/logit; LSTM fed GARCH outputs.

Results	• LSTM beat other methods on direction; edge declined later. • Hybrids reduced volatility errors.
How it relates to MY study	Evidence that LSTMs capture structure; hybrid idea for PhD.
Must read in the original	Introductions and results.
Can skim / skip	—
Key terms	hybrid model
Strengths	Widely cited.
Weaknesses / critique	Confirm details.
Use it as · where	Support · 2.5

Bucci, A. (2020) Realized volatility forecasting with neural networks. JFEC, 18(3), 502–531.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Compares feed-forward and recurrent networks with econometric models for realised volatility.
Dataset	Monthly S&P 500, Feb 1950 – Dec 2017 (815 obs.); log RV from daily squared returns; macro and financial predictors (Table 1).
Methodology	FNN, Elman, Jordan, LSTM, NARX vs ARFIMA(X), LSTAR(X); rolling window; 245 out-of-sample forecasts (1997–2017), 22 in the 2007–09 subsample; MSE and QLIKE; Model Confidence Set; DM tests.
Results	• Recurrent networks outperform traditional econometric methods. • LSTM and NARX are best and beat FNN, Elman, Jordan. • Advantage also in 2007–09.
How it relates to MY study	Main support for H1; but monthly data and only 22 crisis forecasts — your daily/weekly, three-crisis design is a clear improvement.
Must read in the original	Data (pp.4–6); LSTM description (p.11, Figure 6); forecast accuracy (pp.18–26, Tables 4–6 pp.21–22); conclusions (p.27).
Can skim / skip	Other network architectures (pp.7–10, 12–13).
Key terms	NARX; Model Confidence Set; ARFIMA
Strengths	Careful comparison; robust losses.
Weaknesses / critique	Monthly; tiny crisis sample; no explanations.
Use it as · where	Support + contrast · 2.5; H1; gap

Christensen, K., Siggaard, M. & Veliyev, B. (2023) A machine learning approach to volatility forecasting. JFEC, 21(5), 1680–1727.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Large comparison of machine-learning methods with HAR models for realised variance.
Dataset	DJIA constituents, 29 Jan 2001 – 31 Dec 2017 (4,257 days); intraday RV; many predictors (Table 1 p.16).
Methodology	Regularisation, trees, random forest, boosting, neural networks vs HAR family; minimal tuning; 1-day, 1-week, 1-month horizons; rolling HAR; fixed window for neural networks (compute cost); accumulated local effects (ALE) for importance.
Results	• ML competitive and beats HAR even with only HAR lags. • Gains small at 1 day, larger at weekly and monthly horizons. • Random forest and neural networks best. • Models agree on the top predictors but disagree on their ranking.
How it relates to MY study	Supports H1 and your weekly target; ranking disagreement motivates RQ4; fixed-window NN supports your annual refits.
Must read in the original	Introduction (pp.2–3); HAR (pp.4–5); data (Table 1 p.16); 1-day results (Tables 2–3 pp.17–19); ALE and importance (Figures 6–7 pp.25–26); weekly results (Tables 4–5 pp.28–29); conclusion (end).
Can skim / skip	Individual ML algorithm descriptions you already know; appendices.
Key terms	HAR-X; accumulated local effects; relative MSE
Strengths	Broad, careful.
Weaknesses / critique	US large caps; intraday data.
Use it as · where	Support · 2.5–2.6; H1; H4

Bergmeir & Benítez (2012) Information Sciences, 191.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	How to validate forecasting models on time series.
Dataset	Simulations and real series.
Methodology	Compare validation schemes.
Results	• Keep time order; use blocked or forward schemes.
How it relates to MY study	Justifies walk-forward evaluation.
Must read in the original	Introduction and conclusions.
Can skim / skip	—
Key terms	blocked cross-validation

Strengths	Practical.
Weaknesses / critique	Confirm details.
Use it as · where	Method · 2.5; Chapter 3

Brini, A. (2026) Forecasting realized volatility with time series foundation models: a comparison with econometric benchmarks. arXiv 2607.05291 (preprint).	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Tests nine pre-trained ‘foundation’ time-series models, used zero-shot, against eight HAR-family econometric benchmarks for realised-volatility forecasting.
Dataset	VOLARE dataset: 50 assets in equities, foreign exchange and futures; three forecast horizons.
Methodology	Zero-shot forecasts vs HAR, Log-HAR and related models; Model Confidence Set.
Results	• No universal advantage for foundation models. • Only one small model (Tiny Time Mixers) beats the benchmark at every horizon, narrowly. • Much of the short-horizon gain reflects better-scaled forecasts, not better dynamics. • An equal-weight average of TTM and Log-HAR matches the best single model (in the Model Confidence Set for 98–100% of assets).
How it relates to MY study	Up-to-date support for your H1 result: newer and larger ML models still struggle to beat HAR-type benchmarks.
Must read in the original	Abstract, introduction and the main results tables.
Can skim / skip	Descriptions of the individual foundation models.
Key terms	foundation model; zero-shot; Log-HAR; Model Confidence Set
Strengths	Many assets; strong benchmarks; very recent.
Weaknesses / critique	Preprint (not peer-reviewed); no crisis-by-crisis or explanation analysis.
Use it as · where	Evidence · 2.5; Discussion of H1

D5. Section 2.6 — Explaining ML forecasts

Lundberg, S. & Lee, S.-I. (2017) A unified approach to interpreting model predictions. NeurIPS 30.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Unifies six explanation methods and shows SHAP is the unique method with three desirable properties.
Dataset	Theory; computational and user studies.
Methodology	Additive feature attribution; Shapley values.
Results	• Local accuracy, missingness, consistency uniquely define SHAP values. • Kernel SHAP (any model) and Deep SHAP (neural networks) estimators.
How it relates to MY study	Definition of what you compute.
Must read in the original	1 Introduction (p.1); 3 properties (p.4); 4 SHAP values (pp.4–5); 4.1–4.2 approximations (pp.5–8); conclusion (p.9).
Can skim / skip	5 user studies (p.8).
Key terms	additive feature attribution; local accuracy; missingness; consistency
Strengths	Theoretical foundation.
Weaknesses / critique	Estimators assume feature independence.
Use it as · where	Method · 2.6; Chapter 3

Sundararajan, M., Taly, A. & Yan, Q. (2017) Axiomatic attribution for deep networks. ICML.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Proposes Integrated Gradients from two axioms.
Dataset	Image, text, chemistry models.
Methodology	Axioms; path integral of gradients.
Results	• Sensitivity and implementation invariance violated by most earlier methods. • IG satisfies them; simple to implement. • Does not explain interactions.
How it relates to MY study	Basis of gradient-based SHAP; baseline choice matters.
Must read in the original	2 axioms (p.2); 3 method (p.3); 5 applying IG — choosing baselines (p.5); conclusion (p.8).
Can skim / skip	6 applications (pp.6–8).
Key terms	baseline; completeness; path method
Strengths	Axiomatic.
Weaknesses / critique	Baseline-dependent.
Use it as · where	Method · 2.6; Chapter 3

Erion, G. et al. (2021) Expected gradients. Nature Machine Intelligence, 3, 620–631.

Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Introduces expected gradients (IG averaged over data baselines).
Dataset	Images, genomics, health.
Methodology	Expected gradients; attribution priors.
Results	• Satisfies key axioms; improves training when used as a prior.
How it relates to MY study	Justifies Captum GradientShap with real training windows as background.
Must read in the original	Introduction and the expected-gradients formulation in the Methods section.
Can skim / skip	Application experiments.
Key terms	expected gradients; attribution prior
Strengths	Practical, axiomatic.
Weaknesses / critique	Main aim is training.
Use it as · where	Method · 2.6; Chapter 3

Aas, K., Jullum, M. & Løland, A. (2021) Explaining predictions when features are dependent. AI, 298.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Shows Kernel SHAP can be very wrong with correlated features and extends it.
Dataset	Simulated models; real credit data.
Methodology	Dependence-aware Kernel SHAP.
Results	• Independence assumption can give very wrong explanations, even for linear models. • Dependence-aware versions are more accurate.
How it relates to MY study	Justifies keeping only three low-overlap markets (Nasdaq-S&P 0.95, FTSE-DAX 0.84 in your data).
Must read in the original	1 Introduction (pp.1–2); 2 Kernel SHAP (pp.3–6); 3 dependence (pp.6–9); Figure 2 (p.13).
Can skim / skip	Most experiments (pp.9–18).
Key terms	feature dependence; conditional expectation
Strengths	Clear demonstration.
Weaknesses / critique	Heavier computation.
Use it as · where	Caution + method · 2.6; H4

Kumar, I. et al. (2020) Problems with Shapley-value-based explanations. ICML.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Critique of Shapley values as feature importance.
Dataset	Theory.
Methodology	Mathematical and human-centred analysis.
Results	• Mathematical problems (conditional vs interventional value functions; additivity). • Fixes need causal reasoning; poor fit with human explanation goals.
How it relates to MY study	Main ‘limits of SHAP’ source; interpretation rule.
Must read in the original	2.2 (p.2); 3.1 conditional vs interventional (pp.3–5); 3.2 additivity (p.5); 4 human-centric (pp.6–8); conclusion (p.9).
Can skim / skip	—
Key terms	value function; interventional; contrastive explanation
Strengths	Sharp critique.
Weaknesses / critique	No alternative method.
Use it as · where	Contrast · 2.6; Limitations

Rudin, C. (2019) Stop explaining black box models. Nature Machine Intelligence, 1, 206–215.	
Source basis	FULL TEXT read in this session; page numbers refer to the open-access version in Part G
Summary	Argues for inherently interpretable models in high-stakes settings.
Dataset	Examples from criminal justice, healthcare.
Methodology	Position paper.
Results	• Explaining black boxes can perpetuate bad practice; build interpretable models instead.
How it relates to MY study	Counter-argument: your benchmarks are interpretable; SHAP is used for research diagnosis.
Must read in the original	1 Introduction (p.1); 2 key issues with explainable ML (pp.2–6); conclusion (p.15).
Can skim / skip	5 algorithmic challenges (pp.9–15).
Key terms	interpretable vs explainable ML
Strengths	Influential.
Weaknesses / critique	Argumentative.
Use it as · where	Contrast · 2.6; Discussion

Ahmed, A. (2026) LSTM-Attention-SHAP framework for volatility. SSRN preprint 6301119.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Recent preprint combining LSTM, attention and SHAP for volatility.
Dataset	Five asset classes, 2018–2024.
Methodology	LSTM + attention + SHAP.
Results	Reported $\approx 28.6\%$ lower RMSE than GARCH(1,1); SHAP ranks a geopolitical-risk index then VIX (verify).
How it relates to MY study	Closest explainable-volatility work; descriptive SHAP, no cross-crisis test.
Must read in the original	Abstract, method, SHAP results.
Can skim / skip	—
Key terms	attention
Strengths	Recent.
Weaknesses / critique	Not peer-reviewed; RMSE only.
Use it as · where	Contrast · 2.6; 2.7 gap

Feng, L., Shi, J. & Kutun, A.M. (2026) Your fear is (partly) mine: the role of non-VIX volatility in forecasting regional stock market volatility using interpretable machine learning. JIMF, 160, 103467.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Asks how much other markets' volatility indices — not only the US VIX — help forecast a market's volatility, using automated machine learning with interpretation.
Dataset	Five-minute data, 2 Mar 2015 – 1 Mar 2024; four developed markets (US, Germany, Japan, Hong Kong) and two emerging markets (China, India).
Methodology	Dynamic automated machine-learning models; interpretable-ML analysis of each predictor's contribution (check the exact method when you read it).
Results	- Local volatility indices forecast domestic volatility best, especially in stable periods. - The VIX spills over strongly, less so to emerging markets. - After COVID-19, China's volatility index overtook the VIX in predictive role. - Spillovers are asymmetric and stronger in turbulent periods.
How it relates to MY study	The closest recent study to your channel shares: foreign vs domestic contributions to volatility forecasts that change with market conditions. Their 'local dominates in calm periods' parallels your domestic share of about 0.5.
Must read in the original	Abstract; data and model section; the interpretation results (full text via the DBS library).
Can skim / skip	Market-by-market detail beyond the US and Germany.
Key terms	VIX; implied-volatility index; interpretable machine learning; spillover asymmetry
Strengths	Recent; strong journal; cross-market; interpretable.
Weaknesses / critique	Uses implied-volatility indices (no Irish equivalent); only the abstract was read in this session.
Use it as · where	Positioning · 2.6 and 2.7 (must cite)

D6. Methods and evidence of the ML extensions E3–E4 (added 1 Oct 2026)

Breiman, L. (2001) Random forests. Machine Learning, 45(1), 5–32.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Introduces random forests: many decision trees, each grown on a bootstrap sample and choosing among a random subset of inputs at each split; the forecast is the average of the trees.
Dataset	Benchmark data sets and simulations (confirm).
Methodology	Bagging plus random input selection; out-of-bag error estimates; variable importance.
Results	- Accuracy comparable to boosting and robust to noise. - Out-of-bag error gives an honest error estimate without a separate test set. - Adding more trees does not over-fit.
How it relates to MY study	Your E3 Random Forest (taught in the MSc module). A forest's forecast is an average of training targets, so it can never exceed the largest value it has seen — the ceiling behind its GFC failure (2008 peak forecast 21 against an actual 292, $\times 10^{-4}$).
Must read in the original	The definition of the forest, out-of-bag estimates and variable importance (confirm sections in your copy).
Can skim / skip	Convergence and strength–correlation theory.
Key terms	bagging; out-of-bag; max_features; variable importance
Strengths	Few settings; hard to over-fit; works well out of the box.
Weaknesses / critique	Cannot extrapolate beyond the training range; impurity importance is biased when inputs are correlated.
Use it as · where	Method · Chapter 3 (E3); Discussion of the training-range ceiling

Drucker, H., Burges, C., Kaufman, L., Smola, A. & Vapnik, V. (1997) Support vector regression machines. Advances in Neural Information Processing Systems 9, 155–161.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Extends support vector machines to regression: errors smaller than ϵ are ignored, C trades flatness against errors, and

	kernels allow non-linear fits.
Dataset	Simulated test functions and a standard regression data set (confirm).
Methodology	ϵ -insensitive loss; kernel functions; comparison with bagged regression trees.
Results	- SVR did as well as or better than bagged trees. - Only the support vectors determine the fit.
How it relates to MY study	Your E3 SVR. With an RBF kernel, a day unlike any training day gets little weight, so the forecast falls back towards the model's average — your 2008 shock test ($\times 1.2$ at a -20% day).
Must read in the original	The method section (ϵ -insensitive loss, kernels).
Can skim / skip	Experimental details.
Key terms	ϵ -insensitive loss; kernel; C ; gamma; support vectors
Strengths	Handles non-linearity with few settings; regularised.
Weaknesses / critique	Needs scaled inputs; slow on large samples; RBF kernels cannot extrapolate.
Use it as · where	Method · Chapter 3 (E3)

Chen, T. & Guestrin, C. (2016) XGBoost: a scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD Conference (KDD '16), 785–794.

Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Describes XGBoost: gradient boosting of trees, each tree fitted to the errors of the trees before it, with regularisation and fast, scalable computation.
Dataset	Large benchmark data sets (confirm).
Methodology	Regularised objective; second-order approximation; shrinkage (learning rate) and column subsampling; sparsity-aware split finding.
Results	- Strong results in many machine-learning competitions. - Scales to very large data sets.
How it relates to MY study	Your E4 XGBoost failed in the GFC like the Random Forest (QLIKE 0.925; 2008 peak forecast 27 against 292): a sum of leaf values learnt from calm years cannot reach crisis levels. As a GARCH hybrid its GFC QLIKE fell to 0.455.
Must read in the original	Sections 2–3 at a conceptual level (objective, tree boosting, split finding).
Can skim / skip	Systems sections (cache-aware access, out-of-core computation).
Key terms	gradient boosting; learning rate; max_depth; min_child_weight; subsample
Strengths	Strong on tabular data; built-in regularisation.
Weaknesses / critique	Many settings to tune; the tree version cannot extrapolate.
Use it as · where	Method · Chapter 3 (E4)

Corsi, F. & Renò, R. (2012) Discrete-time volatility forecasting with persistent leverage effect and the link with continuous-time volatility modeling. Journal of Business & Economic Statistics, 30(3), 368–380.

Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Adds a leverage effect to the HAR model: past negative returns, averaged over a day, a week and a month, raise future volatility; jumps are modelled as well.
Dataset	US stock-index futures, high-frequency data (confirm the asset and period).
Methodology	HAR regression with negative-return terms $\min(r, 0)$ at three horizons, plus jump terms; link to continuous-time models.
Results	- Leverage is persistent: falls raise volatility for weeks, not only the next day. - Leverage and jump terms improve forecasts.
How it relates to MY study	Source of the leverage terms in your E4. In your HAR-X they entered a log model linearly, and in October 2008 the forecasts exploded (up to $9,955 \times 10^{-4}$) — a specification point for your discussion.
Must read in the original	The model specification and the forecasting results.
Can skim / skip	Continuous-time derivations.
Key terms	leverage effect; LHAR; jumps; heterogeneous horizons
Strengths	Simple extension of HAR; strong empirical support.
Weaknesses / critique	Linear leverage terms in a log model can explode in an unprecedented crash (your E4).
Use it as · where	Method · Chapter 3 (E4); Discussion

Audrino, F. & Knaus, S.D. (2016) Lassoing the HAR model: a model selection perspective on realized volatility dynamics. Econometric Reviews, 35(8–10), 1485–1521.

Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Lets the lasso choose which past volatility lags matter, and asks whether it recovers the HAR structure.
Dataset	Realised volatility series (confirm the assets and period in the paper).
Methodology	Lasso on many lags of realised volatility; theory for the case where HAR is the true model; out-of-sample comparison with HAR.
Results	- If HAR is the true model, the lasso recovers its lag structure in large samples. - In the data the lasso picks a different lag structure from HAR, yet forecasts about as well.
How it relates to MY study	Linear machine learning (penalised regression) does no better than HAR — support for reading your HAR and GARCH results as hard to beat. A LASSO-HAR with many lags is a possible PhD extension.

Must read in the original	Introduction, empirical section and conclusion.
Can skim / skip	Proofs.
Key terms	lasso; lag selection; sparsity; HAR
Strengths	Combines theory and data.
Weaknesses / critique	Only the abstract was read; linear models only.
Use it as · where	Evidence · 2.5

Branco, R.R., Rubesam, A. & Zevallos, M. (2024) Forecasting realized volatility: does anything beat linear models? Journal of Empirical Finance, 78, 101524.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Compares linear models (HAR family and penalised regressions) with non-linear machine-learning models for forecasting the realised volatility of ten stock indices.
Dataset	Ten global stock-market indices, January 2000 – December 2021. Predictors: past realised volatility, lagged returns, implied volatility, macroeconomic and sentiment variables.
Methodology	Out-of-sample forecasts at daily, weekly and monthly horizons; MSE and Value-at-Risk evaluation; Model Confidence Set; volatility-timing portfolios for economic value.
Results	• Extra predictors improve daily and weekly forecasts. • No evidence that non-linear ML beats linear models in general. • All ML models tend to under-forecast in high-volatility periods and over-forecast in calm ones. • Simpler models give more economic value in monthly volatility-timing portfolios.
How it relates to MY study	The closest large-sample match to your E3–E4: non-linear learners do not beat linear benchmarks, and they under-forecast when volatility is high (your learners under-forecast on 69% of GFC days). Implied volatility helps, as in your HAR-X with the VIX.
Must read in the original	Data and model list; main results tables; the high- vs low-volatility analysis (full text via the DBS library).
Can skim / skip	Descriptions of the individual ML algorithms.
Key terms	HARX; Model Confidence Set; volatility timing; implied volatility
Strengths	Ten indices over 22 years; many models; statistical and economic evaluation.
Weaknesses / critique	Only the abstract was read (working-paper and journal versions); check whether crises are analysed separately.
Use it as · where	Evidence · 2.5; Discussion of H1, E3 and E4 (must cite)

Audrino, F. & Chassot, J. (2024) HARd to beat: the overlooked impact of rolling windows in the era of machine learning. arXiv 2406.08041 (preprint).	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Shows that the length of the training window and how often a model is re-estimated matter more than the choice between HAR and machine learning.
Dataset	1,445 US stocks; daily realised volatility from 5-minute TAQ data; January 2016 – November 2023 (test period January 2022 – November 2023); the VIX as an extra predictor.
Methodology	HAR (including weighted least squares) with rolling windows of 250–2,000 days and re-estimation every 1–250 days, against lasso, random forest, gradient-boosted trees and neural networks; one-day horizon; MSE, QLIKE and realised utility; Model Confidence Set.
Results	• Daily re-estimation matters: accuracy falls even when re-estimating every 2–5 days. • HAR estimated by weighted least squares beats every ML model on MSE, QLIKE and utility. • Best windows are about 2.5–4 years. • Machine learning is 'not a panacea' when the inputs are past volatility and the VIX.
How it relates to MY study	Every model in your study is re-estimated once a year (pre-registered). This paper warns that the refit frequency can change rankings — a limitation to state, and a cheap PhD extension.
Must read in the original	Section 4.3 (estimation strategy) and Section 5 (results).
Can skim / skip	Descriptions of the ML algorithms.
Key terms	rolling window; re-estimation frequency; HAR-WLS; realised utility
Strengths	Very large sample; systematic design; several losses.
Weaknesses / critique	Preprint (check for a published version); short test period (2022–23); one-day horizon only.
Use it as · where	Evidence · 2.5; Limitations

Teller, A., Pigorsch, U. & Pigorsch, C. (2022) Short- to long-term realized volatility forecasting using extreme gradient boosting. SSRN working paper 4267541.	
Source basis	ABSTRACT / summary read in this session; read the full paper before relying on details
Summary	Uses XGBoost to forecast realised volatility at short and longer horizons, against HAR and LSTM models.
Dataset	Realised volatility (confirm the assets and period in the paper).
Methodology	XGBoost with linear and with tree base learners vs HAR and LSTM; several forecast horizons.
Results	• XGBoost beats HAR and LSTM one step ahead. • At longer horizons XGBoost with linear base learners does better than the tree version: non-linearity matters mainly for short horizons.

How it relates to MY study	Your horizon is 5 days. Their result fits your finding that the linear and structural models do best and that tree models add little.
Must read in the original	Abstract and the results by horizon.
Can skim / skip	—
Key terms	XGBoost; linear base learner; forecast horizon
Strengths	Direct comparison of XGBoost, HAR and LSTM.
Weaknesses / critique	Working paper, not peer-reviewed; only the abstract was read.
Use it as · where	Evidence · 2.5 (cite as a working paper)

Bates, J.M. & Granger, C.W.J. (1969) The combination of forecasts. Operational Research Quarterly, 20(4), 451–468.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	Shows that combining two forecasts can be more accurate than either, with weights based on their past errors.
Dataset	Airline passenger forecasts (confirm).
Methodology	Weighted combinations of two forecasts; weights from past error variances.
Results	- The combination often has a smaller error than either forecast. - Simple weights work well.
How it relates to MY study	Your E4 average of GARCH and LSTM matched GARCH overall (QLIKE 0.380), was better in COVID-19, 2022 and calm days, and worse in the GFC.
Must read in the original	The main argument and the worked examples.
Can skim / skip	—
Key terms	forecast combination; combination weights; forecast-combination puzzle
Strengths	Foundational; simple.
Weaknesses / critique	Old; assumes the forecasts' error properties are stable.
Use it as · where	Method · Chapter 3 (E4); Discussion

Holm, S. (1979) A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics, 6(2), 65–70.	
Source basis	STANDARD KNOWLEDGE, not re-read this session; confirm every detail and page before citing
Summary	A step-down method that limits the chance of any false positive when several tests are run together.
Dataset	— (statistical method).
Methodology	Sort the p-values; compare the smallest with α/m , the next with $\alpha/(m-1)$, and so on; stop at the first test that is not rejected.
Results	- Controls the family-wise error rate. - Always at least as powerful as the Bonferroni correction.
How it relates to MY study	Used for the primary family of your A3 (three GARCH hybrids vs GARCH): Holm $p = 1.0, 1.0$ and 0.36 .
Must read in the original	The procedure (a short paper).
Can skim / skip	—
Key terms	family-wise error rate; Bonferroni; multiple testing
Strengths	Simple; no assumption about how the tests depend on each other.
Weaknesses / critique	Conservative when there are many correlated tests.
Use it as · where	Method · Chapter 3 (E4)

E. Reading order (about 32 hours in total)

Week	Read (sections from the cards)	Hours
1	Minsky (all); Reinhart & Rogoff pp.2–15; Fama/Cont summaries; Kelly	4
1–2	Brunnermeier pp.4–30; Bekaert pp.8–12, 20–26; Acharya pp.2–4, 22–40; official report summaries	6
2	Corbet & Twomey (all); Lane; Whelan	3
3	Forbes & Rigobon pp.3–16, 22–24, 36–37; Diebold & Yilmaz pp.3–10; Engle et al.; Hamao	4
3–4	Patton pp.1–8; Corsi; Hansen & Lunde; Bucci pp.4–6, 18–27; Christensen pp.2–5, 16–29	5
4–5	Lundberg & Lee pp.1–9; Sundararajan pp.2–5; Aas pp.1–9; Kumar pp.2–9; Rudin pp.1–6	3
Now	Diebold & Yilmaz (2012) pp.58–61; Danielsson et al. (2018) pp.3–6, 13–14, 19–22; Pesaran & Shin (1998); Feng et al. (2026); Brini (2026)	3
Now (A2–A3)	Branco et al. (2024) data and results; Audrino & Chassot (2024) Sections 4.3 and 5; Corsi & Renò (2012) model section; Breiman (2001) definition and out-of-bag; Bates & Granger (1969)	4

F. From notes to your own paragraph

Part	What it does	Planning example (bullet notes, not text to copy)
1 Point	Your claim	GFC reached Ireland through both a US and a domestic channel
2 Evidence	Two or more sources	Bekaert et al. (domestic contagion, p.25); Kelly (7×/11× lending); Figure H1 (ISEQ fell further than others after Lehman)
3 Contrast	A complicating source	Reinhart & Rogoff — crises start in financial centres
4 Critique	Limits of the evidence	Factor-model definition; Forbes & Rigobon volatility bias
5 So what	Link to your study	Expect US and domestic shares to rise in the GFC window (H2a)

G. Where to read the originals

Source	Open-access version used for page numbers
Minsky (1992)	levyinstitute.org/pubs/wp74.pdf
Reinhart & Rogoff	nber.org/papers/w13882
Brunnermeier	nber.org/papers/w14612
Bekaert et al.	ecb.europa.eu/pub/pdf/scpwps/ecbwp1381.pdf
Acharya et al.	pages.stern.nyu.edu/~pschnabl/research/AcharyaDrechslerSchnabl_final.pdf
Baker et al.	nber.org/papers/w26945
Akhtaruzzaman et al.	ideas.repec.org/a/eee/finlet/v38y2021ics1544612320305754.html
Li, Ahmed & Chevapatrakul	nottingham.ac.uk/cfc/news/papers/1606.aspx
Forbes & Rigobon	nber.org/papers/w7267
Engle, Ito & Lin	nber.org/papers/w2609
Diebold & Yilmaz (2009)	nber.org/papers/w13811
Patton (2011)	public.econ.duke.edu/~ap172/Patton_vol_proxies_JoE_2011.pdf
Bucci (2020)	mpra.ub.uni-muenchen.de/95443/
Christensen et al.	arxiv.org/abs/2601.13014
Hochreiter & Schmidhuber	bioinf.jku.at/publications/older/2604.pdf
Lundberg & Lee	arxiv.org/abs/1705.07874
Sundararajan et al.	arxiv.org/abs/1703.01365
Erion et al.	arxiv.org/abs/1906.10670
Aas et al.	arxiv.org/abs/1903.10464
Kumar et al.	arxiv.org/abs/2002.11097
Rudin	arxiv.org/abs/1811.10154
Kelly; Whelan	ideas.repec.org/p/ucn/wpaper/200932.html ; ideas.repec.org/p/ucn/wpaper/201306.html
Lane; Brunnermeier (journal)	aeaweb.org (JEP is free to read)
Corbet & Twomey	ideas.repec.org/a/prg/jnlefa/v2015y2015i3id143p15-34.html
Sahiner; Li (2020); Boungou & Yatié	via DBS library / IDEAS
Diebold & Yilmaz (2012)	sas.upenn.edu/~fdiebold/papers/paper99/DirectionOfSpillovers_Mar10.pdf (journal version)
Danielsson et al. (2018)	eprints.lse.ac.uk/91136 (accepted version; page numbers above refer to it)
Gallagher & Twomey (1998)	The Economic and Social Review 29(4) — via the DBS library or the ESR archive
Pesaran & Shin (1998)	Economics Letters — via the DBS library
Brini (2026)	arxiv.org/abs/2607.05291
Feng, Shi & Kutun (2026)	ideas.repec.org/a/eee/jimfin/v160y2026ics0261560625002025.html (full text via the DBS library)
Branco et al. (2024)	doi.org/10.1016/j.jempfin.2024.101524 (working paper: papers.ssrn.com/sol3/papers.cfm?abstract_id=4445161)
Audrino & Chassot (2024)	arxiv.org/abs/2406.08041
Teller, Pigorsch & Pigorsch (2022)	papers.ssrn.com/sol3/papers.cfm?abstract_id=4267541
Audrino & Knaus (2016)	Econometric Reviews 35(8–10) — via the DBS library
Corsi & Renò (2012)	Journal of Business & Economic Statistics 30(3) — via the DBS library
Breiman (2001)	Machine Learning 45(1) — via the DBS library
Chen & Guestrin (2016)	arxiv.org/abs/1603.02754
Drucker et al. (1997)	NIPS 9 proceedings — papers.nips.cc
Bates & Granger (1969)	Operational Research Quarterly 20(4) — JSTOR via the DBS library
Holm (1979)	Scandinavian Journal of Statistics 6(2) — JSTOR via the DBS library

Journal versions are paginated differently from working papers: when you cite a page, use the version you actually read, and prefer the published journal version for the reference list.